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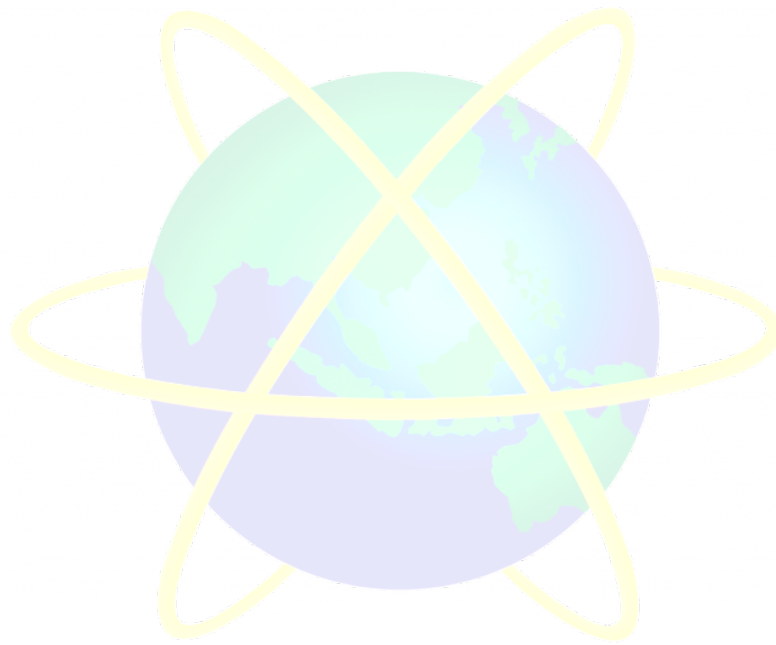
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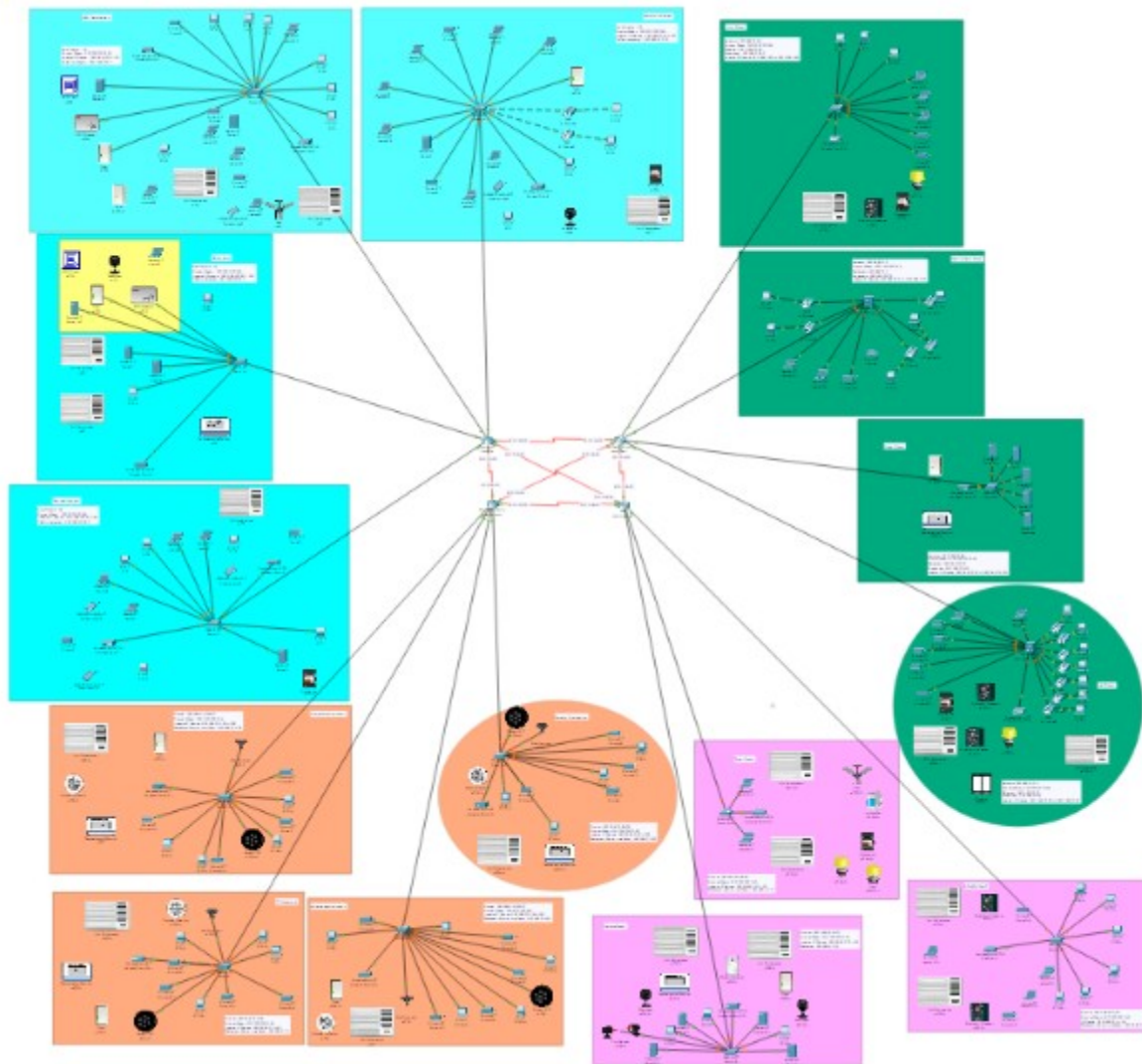


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1.0 Introduction: Muhammad Wijdan(Kuala Lumpur)

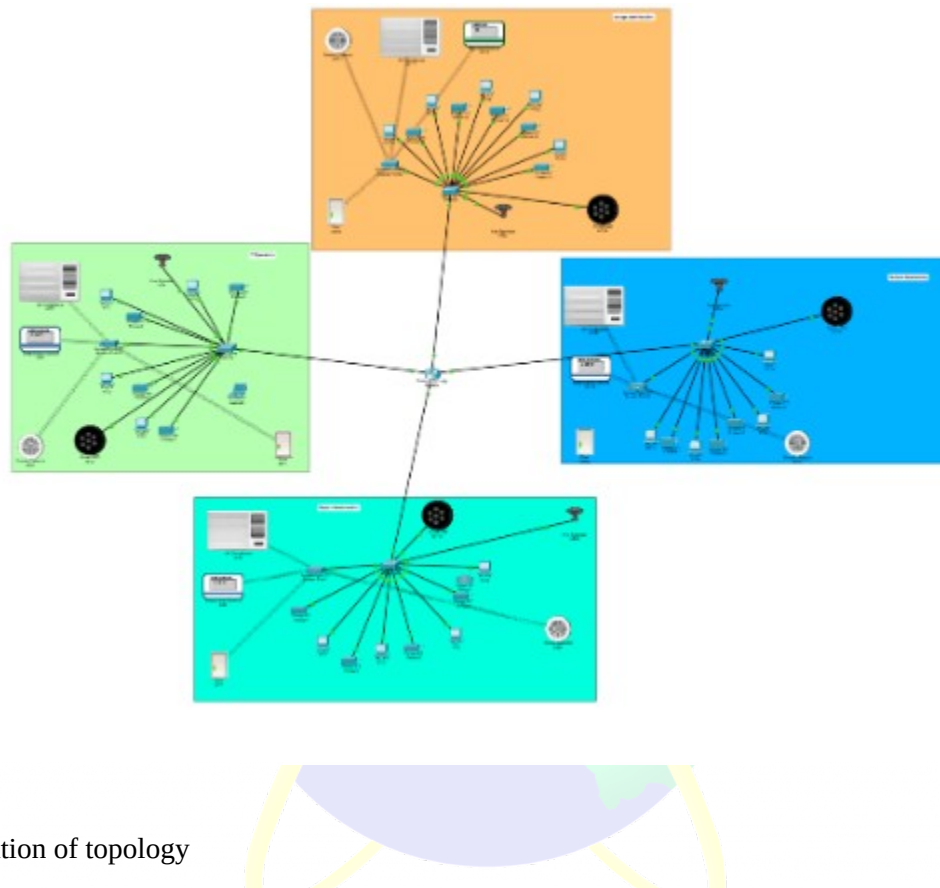
In this networking exercise, a fictional technology company, CodeLink Solutions will be used with the company having four main Malaysian locations based in Kuala Lumpur, Penang, Kota Kinabalu and Johor Bahru. CodeLink Solutions engages in software development, remote technical support, and IT consulting that require a solid and secure network architecture to support daily operations of the organization.

The work involves designing and setting up network structures in each branch so that communication, information sharing, and business continuity in different areas are done well. The solution should allow real-time cooperation, provide safe file transfer facilities, and access centralized resources. Addressing scheme configuration, routing strategies, and simple security installations have been detailed to customise to The requirements of the company given below.



2.0 Student 1: Muhammad Wijdan(Kuala Lumpur)

Figure 2.1 (KL Branch)



2.2 Justification of topology

Network Design Choices for CodeLink Solutions (Kuala Lumpur Branch)

At the Kuala Lumpur office, CodeLink Solutions has well organized local area network (LAN) which allows smooth running of daily activities of software development, as well as the technical support as well as the administrative functions. It uses a clean star topology, which is used in an attempt to satisfy the requirements of the branch set up, which involves centralized control, device availability, high network reliability. In that regard, the structure will enable the branch in Kuala Lumpur to operate independently but with strong links to the central system of the corporation.

Why This Overall Design?

A specialized router connects the branch to the main network, which in turn connects it to other branches via a wide area network (WAN). This makes it possible for secure file sharing, real-time collaboration, and centralized access to company-wide resources. This design facilitates both internal operations and remote coordination by arranging the network with a local server, specialized firewall, and dependable routing. Project syncing and cooperative development work with teams in different places are also made possible by the WAN connectivity.

Why a "Star" Shape in This Branch Network?

A star topology was chosen for the Kuala Lumpur branch to ensure all devices are centrally connected through a single switch. This makes it easier to manage and isolate issues, as a failure in one device does not impact others. The star layout allows for quick reconfiguration and scalability—key features for a growing tech company like CodeLink. This layout ensures that performance remains stable even with multiple users and high bandwidth tasks, such as code deployment or file syncing.

Key Pieces of the Network and What They Do for CodeLink:

The Kuala Lumpur branch includes essential devices such as seven PCs for staff, an IP phone for VoIP communication, a printer for internal documentation, a wireless access point for flexible connectivity, and a local server for file hosting. A firewall is positioned between the internal network and router to secure traffic and prevent unauthorized access. Devices such as CCTV cameras and air conditioning units are also included to reflect real-world office operations and support employee comfort and safety.

What This Means for CodeLink Solutions:

With this setup, the Kuala Lumpur branch is fully capable of handling daily operations independently while maintaining strong connectivity with other locations. The network is designed for speed, security, and flexibility, aligning with the needs of a tech-focused environment. The successful integration of key components ensures reliable performance, data protection, and seamless collaboration, helping CodeLink maintain efficiency and productivity at its Kuala Lumpur site.

2.3 IP Addressing Scheme

2.3.1 Ip Addressing Table

Department	Device Name	Interface	Network Address	IP Address	Default Gateway
IT operations	PC0	Fa0		192.168.20.10	192.168.20.1
	PC1	Fa0		192.168.20.14	
	PC2	Fa0		192.168.20.3	
	PC3	Fa0		192.168.20.4	
	Printer0	Fa0		192.168.20.11	
	Printer1	Fa0		192.168.20.5	

	Printer2	Fa0	192.168.20.0	192.168.20.13	
	Printer3	Fa0		192.168.20.15	
	Smart LED	Fa0		192.168.20.12	
	Fire Sprinkler	Fa0		192.168.20.2	
	Door	Wireless		192.168.20.7	
	Temperature monitor	Wireless		192.168.20.9	
	AC	Wireless		192.168.20.1	
	Smoke detector	Wireless		192.168.20.8	
Storage administration	PC12	Fa0	192.168.20.191	192.168.20.172	192.168.20.161
	PC13	Fa0		192.168.20.168	
	PC14	Fa0		192.168.20.174	
	PC15	Fa0		192.168.20.166	
	PC16	Fa0		192.168.20.170	
	Printer12	Fa0		192.168.20.169	
	Printer13	Fa0		192.168.20.171	
	Printer14	Fa0		192.168.20.173	
	Printer15	Fa0		192.168.20.175	
	Printer16	Fa0		192.168.20.167	
	Fire Sprinkler	Wireless		192.168.20.177	
	Smart LED	Wireless		192.168.20.176	
	Door	Wireless		192.168.20.163	
	Smoke Detector	Wireless		192.168.20.161	
	AC	Wireless		192.168.20.164	
	Humiture Monitor	Wireless		192.168.20.162	
Network Engineering	PC8	Fa0	192.168.20.64	192.168.20.69	192.168.20.65
	PC9	Fa0		192.168.20.74	
	PC10	Fa0		192.168.20.76	

	PC11	Fa0		192.168.20.71	
	Printer8	Fa0		192.168.20.70	
	Printer9	Fa0		192.168.20.72	
	Printer10	Fa0		192.168.20.73	
	Printer11	Fa0		192.168.20.75	
	Smart LED	Fa0		192.168.20.78	
	Fire Sprinkler	Fa0		192.168.20.77	
	Door	Wireless		192.168.20.79	
	Temperature monitor	Wireless		192.168.20.68	
	AC	Wireless		192.168.20.67	
	Smoke Detector	Wireless		192.168.20.65	
System Administration	PC4	Fa0	192.168.20.128	192.168.20.132	192.168.20.129
	PC5	Fa0		192.168.20.130	
	PC6	Fa0		192.168.20.137	
	PC7	Fa0		192.168.20.138	
	Printer4	Fa0		192.168.20.131	
	Printer5	Fa0		192.168.20.139	
	Printer6	Fa0		192.168.20.140	
	Printer7	Fa0		192.168.20.141	
	Smart LED	Fa0		192.168.20.142	
	Fire Sprinkler	Fa0		192.168.20.143	
	AC	Wireless		192.168.20.135	
	Temperature Monitor	Wireless		192.168.20.134	
	Smoke Detector	Wireless		192.168.20.133	
	Door	Wireless		192.168.20.136	

2.4 VLSM Table

Subnet Description	Number of Hosts Needed	Block Size	Subnet mask (in decimal)	prefix	Network address	Broadcast address	Range of usable IP addresses
IT operations	60	64	255.255.255.192	/26	192.168.20.0	192.168.20.63	192.168.20.1-192.168.20.62
Network Engineering	35	64	255.255.255.192	/26	192.168.20.64	192.168.20.127	192.168.20.65-192.168.20.126
System Administration	30	32	255.255.255.224	/27	192.168.20.128	192.168.20.159	192.168.20.129-192.168.20.158
Storage Administration	25	32	255.255.255.224	/27	192.168.20.160	192.168.20.191	192.168.20.161-192.168.20.190



1.0 Student 2: SALEH MOHAMMED SALEH HASAN(Johor Bahru)

1.1 Logical Topology Design

1.1.1 Network Topology Design

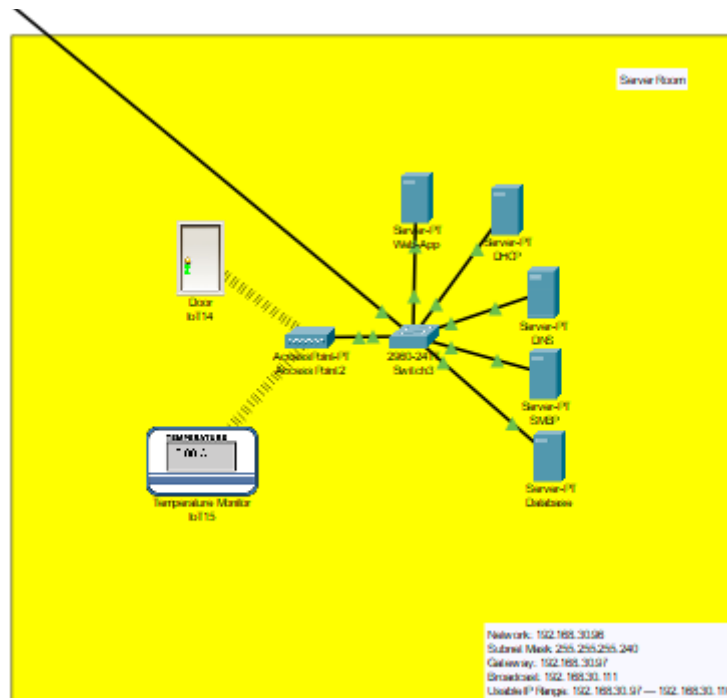


Figure 1.1 JH Server room

Network: 192.168.30.96

Subnet Mask: 255.255.255.240

Gateway: 192.168.30.97

Broadcast: 192.168.30.111

Usable IP Range: 192.168.30.97 — 192.168.30.110

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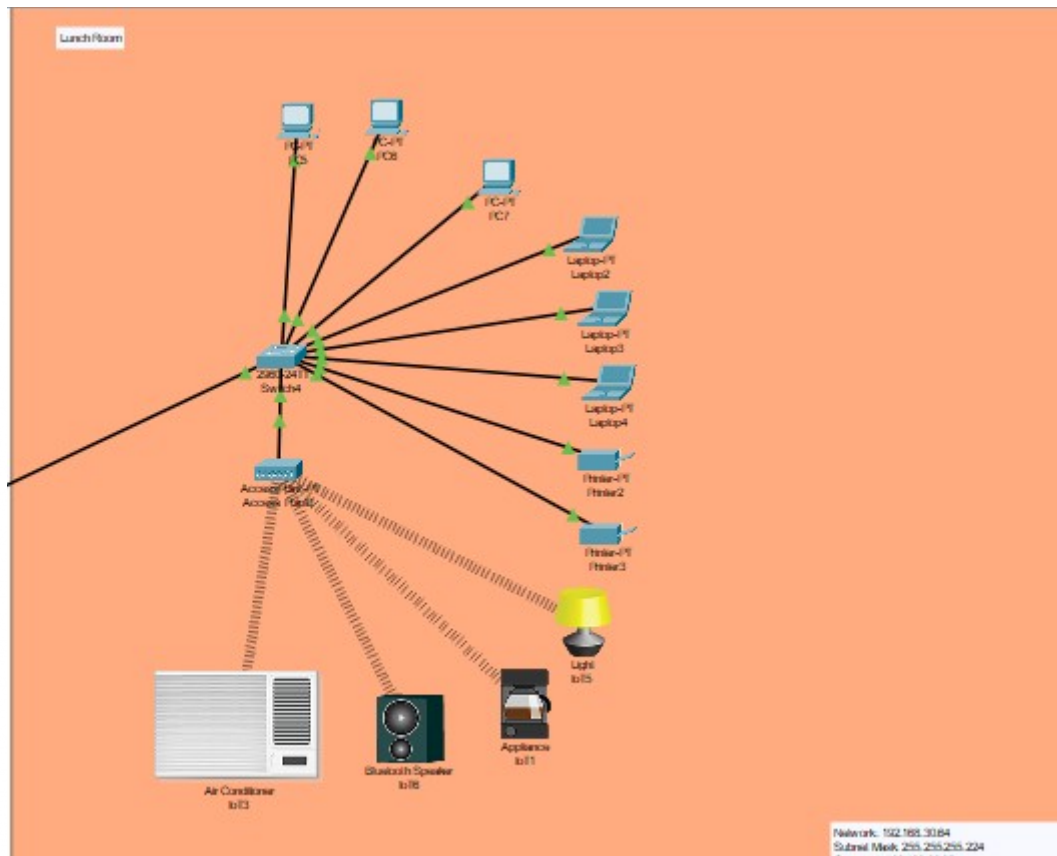


Figure 1.2 KL Lunch room

Network: 192.168.30.64

Subnet Mask: 255.255.255.224

Gateway: 192.168.30.65

Broadcast: 192.168.30.95

Usable IP Range: 192.168.1.65 — 192.168.1.94

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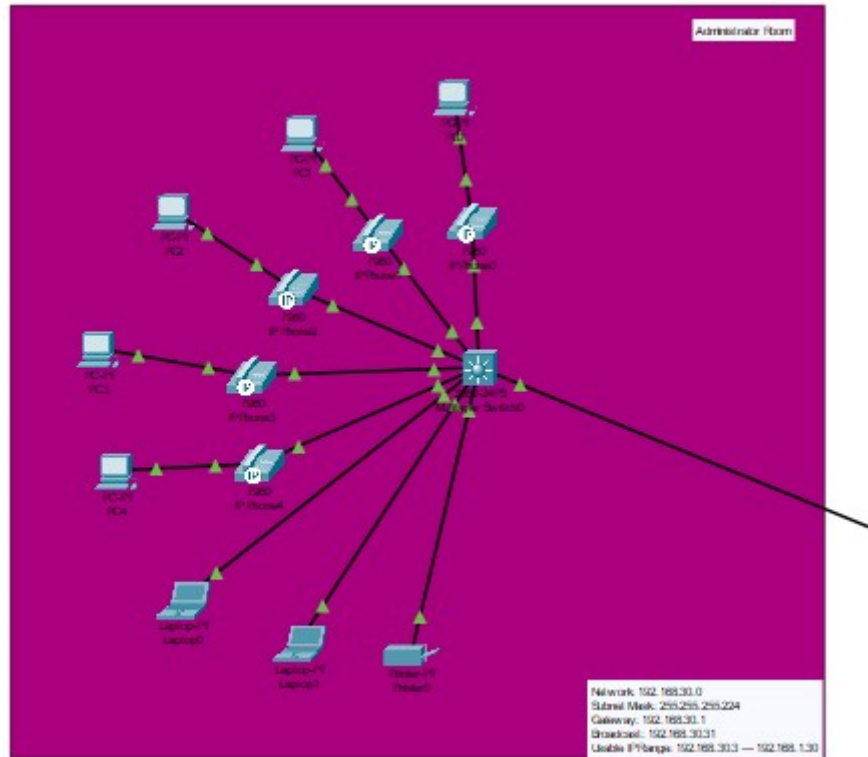


Figure 1.3 KL Admin room

Network: 192.168.30.0

Subnet Mask: 255.255.255.224

Gateway: 192.168.30.1

Broadcast: 192.168.30.31

Usable IP Range: 192.168.30.3 — 192.168.30.30

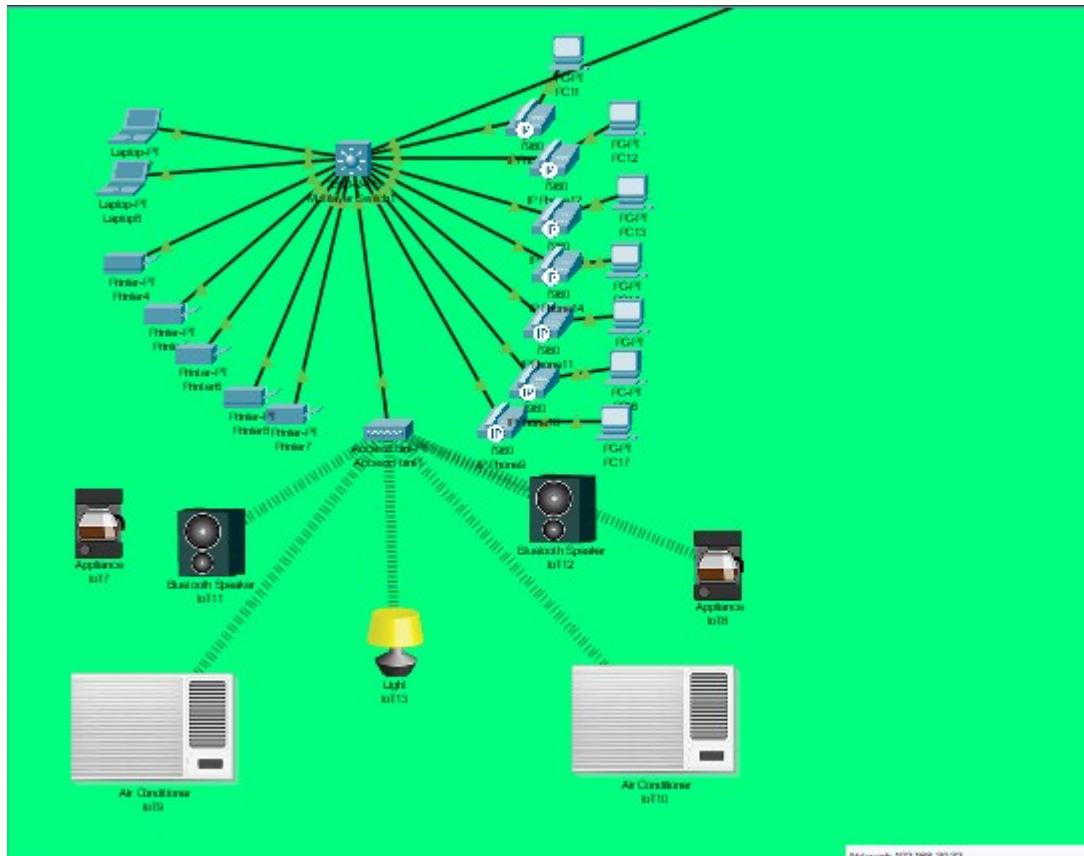


Figure 1.4 Staff room

Network: 192.168.30.32

Subnet Mask: 255.255.255.224

Gateway: 192.168.30.32

Broadcast: 192.168.30.63

Usable IP Range: 192.168.30.65 — 192.168.30.62

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2.2 Topology Design – Introduction

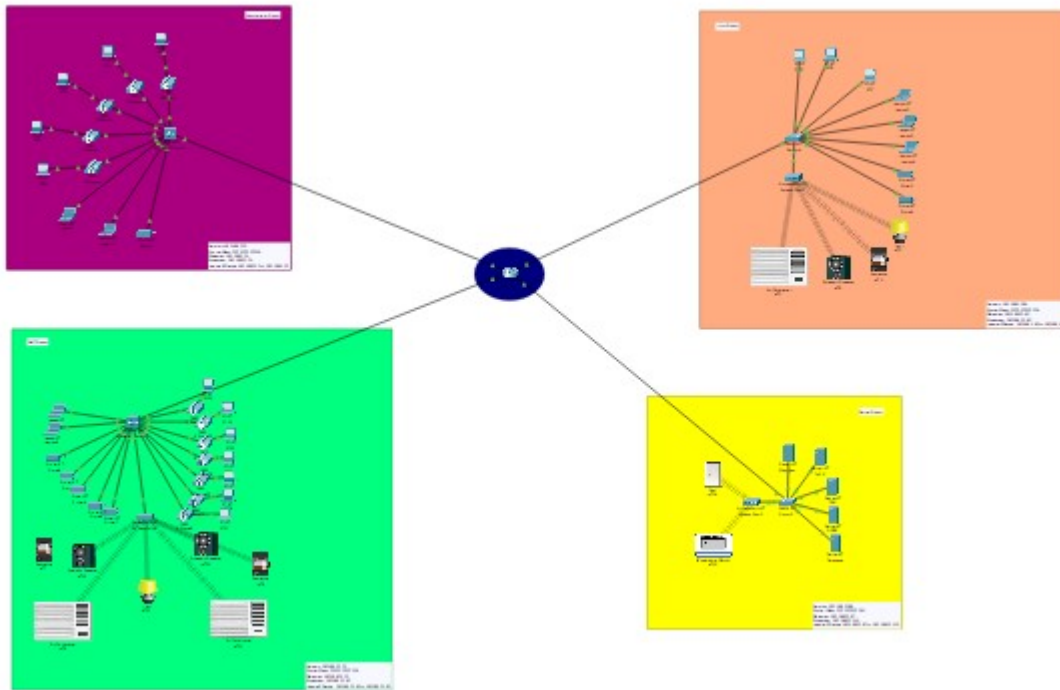


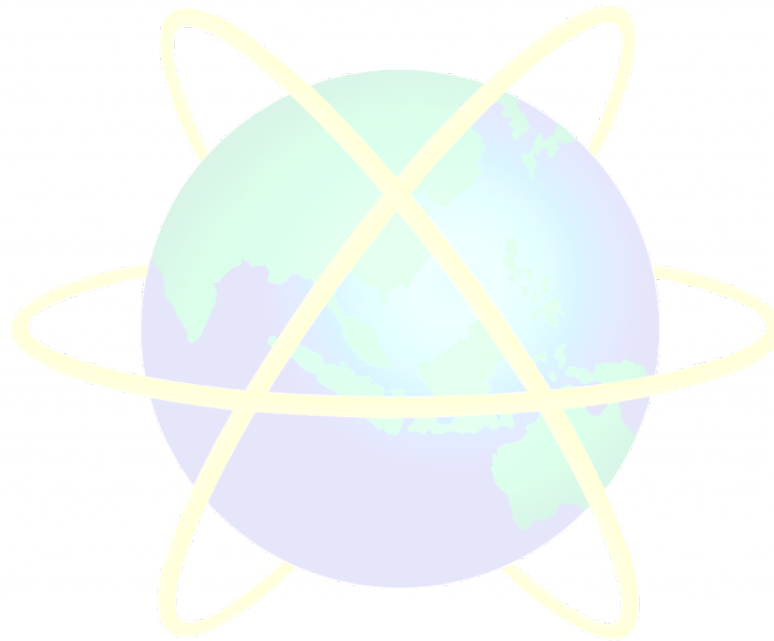
Figure 4.5 KL all departments

In the current business landscape, designing a well-structured and adaptable network topology is essential to facilitate effective, secure, and stable data communication among various organizational sites. This project demands that SkyNetix—a national business expanding across Kuala Lumpur, Penang, Johor Bahru, and Kota Kinabalu—deploy a comprehensive network infrastructure upgrade. This will be used to sustain its growing operations, improve inter-branch collaboration, and accommodate innovative technologies, including Internet of Things (IoT) devices.

The primary goal of this topology design is to establish a fully integrated and operational network architecture that fulfills the operational and technical demands of each branch while providing flawless connectivity via a Wide Area Network (WAN). Every branch should have its own Local Area Network (LAN) infrastructure, tailored to accommodate the individual demands of its departments, including administrative offices, employee workstations, server rooms, and IoT administration areas.

This paper presents the suggested logical topology designs for each of the branches, discussing the reasons for the choice of particular LAN and WAN topologies, along with an IP addressing scheme that facilitates effective address allocation, network partitioning, and future expandability. The primary routing mechanism will involve static routing, taking into account the network's manageable size and the requirement for secure routing paths.

Through the adoption of this ordered and optimized network architecture, SkyNetix seeks to achieve maximum network efficiency, secure data transmission, and establish a stable foundation for business development and future technology advancements.



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2.3 JUSTIFICATION OF THE TOPOLOGY

2.3.1 STAR TOPOLOGY

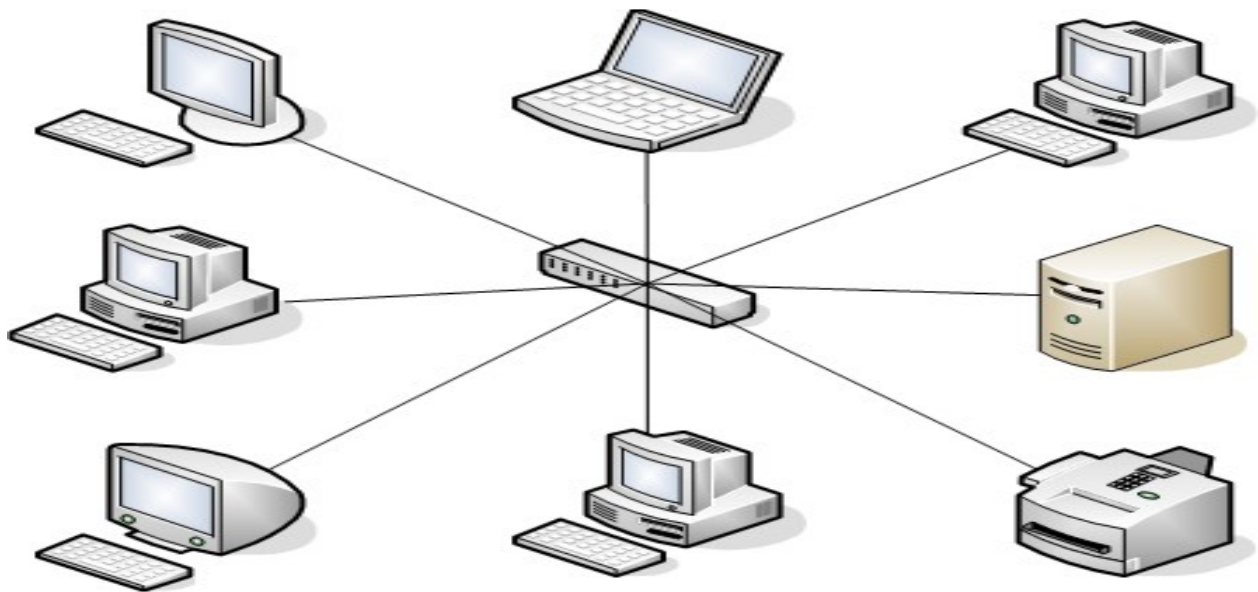


Figure 1.5 Star Topology

Star topology is the best choice for the Kuala Lumpur branch of MapleTech. This option is preferred because it offers simplicity, reliability, scalability, good performance and cost effectiveness (Wevolver, 2023)

The star topology stands out as the network configuration to set up and manage. In this topology each device connects to a switch, which makes troubleshooting problems and adding or removing devices a piece of cake. Moreover, the reliability of the star topology shines through as any device failure does not impact the network or any other hosts (Javatpoint, 2023). Additionally, when it comes to scaling up their network in the future MapleTech can easily expand it by adjusting new devices on the network.

Furthermore, for networks including big or medium sized businesses, like MapleTechs operation this configuration delivers good performance. If necessary though performance can be further enhanced by employing higher quality switches and cables. Lastly implementing a star topology proves less expensive compared to any alternative's topologies.

2.4 IP ADDRESSING SCHEME

2.4.1 IP ADDRESSING TABLE

Device	Interface	Network Address	IP Address	Usable Address Range	Broadcast Address	Default Gateway
Router johor bahru	Gig6/0	192.168.30.64	192.168.30.65	192.168.30.65 — 192.168.30.97	192.168.30.95	N/A
	Gig7/0	192.168.30.96	192.168.30.97	192.168.30.97 — 192.168.40.110	192.168.30.111	N/A
	Gig8/0	192.168.30.32	192.168.30.33	192.168.30.33 — 192.168.30.62	192.168.30.63	N/A
	Gig9/0	192.168.30.0	192.168.30.1	192.168.30.1 - 192.168.30.30	192.168.40.31	N/A
Administration Switch	Gig0/1	192.168.30.0	192.168.30.2	192.168.30.3 — 192.168.1.30	192.168.40.31	192.168.30.1
Lunch Switch	Gig0/1	192.168.30.64	192.168.30.66	192.168.1.65 — 192.168.1.94	192.168.30.95	192.168.30.65
Staff switch	Gig0/1	192.168.30.32	192.168.30.34	192.168.30.65 — 192.168.30.62	192.168.30.63	192.168.30.33
Server switch	Gig0/1	192.168.30.96	192.168.30.98	192.168.30.97 — 192.168.30.110	192.168.30.111	192.168.30.97
PC 0	Fa0	192.168.30.0	192.168.30.3	N/A	N/A	192.168.30.1
PC 1		192.168.30.	192.168.30.	N/A	N/A	192.168.30.

		0	4			1
PC 2		192.168.30. 0	192.168.30. 5	N/A	N/A	192.168.30. 1
PC 3		192.168.30. 0	192.168.30. 6	N/A	N/A	192.168.30. 1
PC4		192.168.30. 0	192.168.30. 7	N/A	N/A	192.168.30. 1
Laptop 0		192.168.30. 0	192.168.30. 8	N/A	N/A	192.168.30. 1
Laptop 1		192.168.30. 0	192.168.30. 9	N/A	N/A	192.168.30. 1
Printer 0		192.168.30. 0	192.168.30. 10	N/A	N/A	192.168.30. 1
PC5		192.168.30. 64	192.168.30. 77	N/A	N/A	192.168.30. 65
PC6		192.168.30. 64	192.168.30. 68	N/A	N/A	192.168.30. 65
PC7		192.168.30. 64	192.168.30. 69	N/A	N/A	192.168.30. 65
Laptop 2		192.168.30. 64	192.168.30. 70	N/A	N/A	192.168.30. 65
Laptop 3		192.168.30. 64	192.168.30. 71	N/A	N/A	192.168.30. 65
Laptop 4		192.168.30. 64	192.168.30. 72	N/A	N/A	192.168.30. 65
Printer 2		192.168.30. 64	192.168.30. 73	N/A	N/A	192.168.30. 65
Printer 3		192.168.30. 64	192.168.30. 74	N/A	N/A	192.168.30. 65
Light		192.168.30. 64	192.168.30. 75	N/A	N/A	192.168.30. 65
Coffe machine		192.168.30. 64	192.168.30. 76	N/A	N/A	192.168.30. 65

Speaker		192.168.30. 64	192.168.30. 78	N/A	N/A	192.168.30. 65
AC		192.168.30. 64	192.168.30. 79	N/A	N/A	192.168.30. 65
PC11		192.168.30. 32	192.168.30. 55	N/A	N/A	192.168.30. 32
PC12		192.168.30. 32	192.168.30. 35	N/A	N/A	192.168.30. 32
PC13		192.168.30. 32	192.168.30. 36	N/A	N/A	192.168.30. 32
PC14		192.168.30. 32	192.168.30. 37	N/A	N/A	192.168.30. 32
PC15		192.168.30. 32	192.168.30. 38	N/A		192.168.30. 32
PC16		192.168.30. 32	192.168.30. 39	N/A		192.168.30. 32
PC17		192.168.30. 32	192.168.30. 40	N/A	N/A	192.168.30. 32
Laptop7		192.168.30. 32	192.168.30. 41	N/A	N/A	192.168.30. 32
Laptop8		192.168.30. 32	192.168.30. 42	N/A	N/A	192.168.30. 32
Printer4		192.168.30. 32	192.168.30. 43	N/A	N/A	192.168.30. 32
Printer5		192.168.30. 32	192.168.30. 44	N/A	N/A	192.168.30. 32
Printer6		192.168.30. 32	192.168.30. 45	N/A	N/A	192.168.30. 32
Printer7		192.168.30. 32	192.168.30. 46	N/A	N/A	192.168.30. 32
Printer8		192.168.30. 32	192.168.30. 47	N/A	N/A	192.168.30. 32
Speaker		192.168.30.	192.168.30.	N/A	N/A	192.168.30.

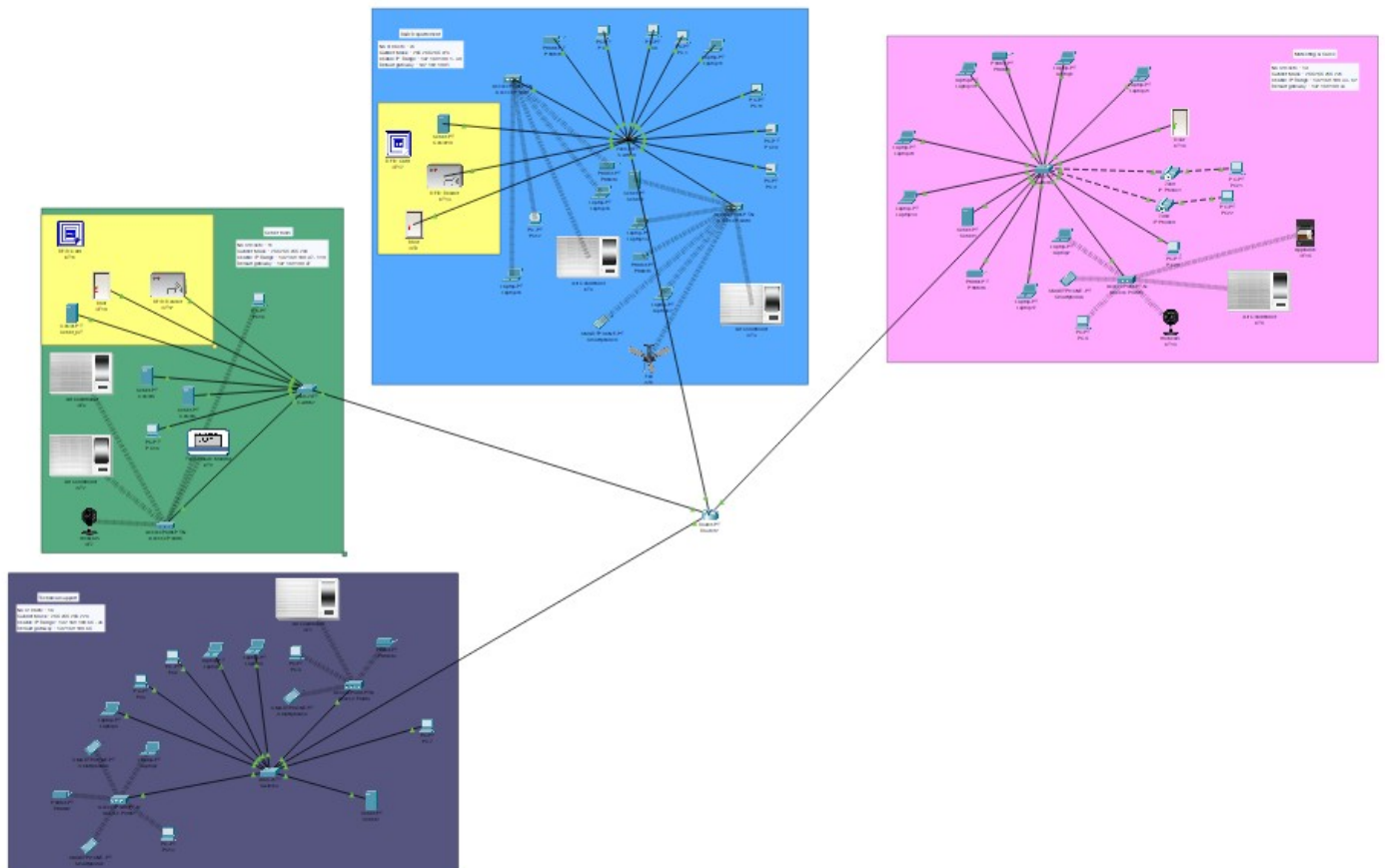
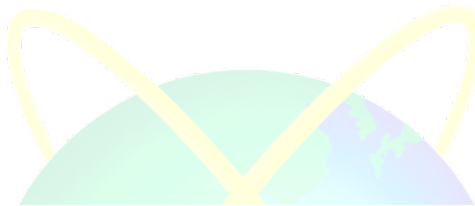
		32	48			32
Speaker		192.168.30.32	192.168.30.49	N/A	N/A	192.168.30.32
Coffee		192.168.30.32	192.168.30.50	N/A	N/A	192.168.30.32
Coffee		192.168.30.32	192.168.30.51	N/A	N/A	192.168.30.32
Light		192.168.30.32	192.168.30.52	N/A	N/A	192.168.30.32
AC		192.168.30.32	192.168.30.53	N/A	N/A	192.168.30.32
Web-App server		192.168.30.96	192.168.30.99	N/A	N/A	192.168.30.97
DHCP server		192.168.30.96	192.168.30.100	N/A	N/A	192.168.30.97
DNS server		192.168.30.96	192.168.30.101	N/A	N/A	192.168.30.97
SMBP server		192.168.30.96	192.168.30.102	N/A	N/A	192.168.30.97
Database server		192.168.30.96	192.168.30.103	N/A	N/A	192.168.30.97
Door		192.168.30.96	192.168.30.104	N/A	N/A	192.168.30.97
Temperature monitor		192.168.30.96	192.168.30.105	N/A	N/A	192.168.30.97

1.0 Introduction : - Penang Branch – OMER MAHDI

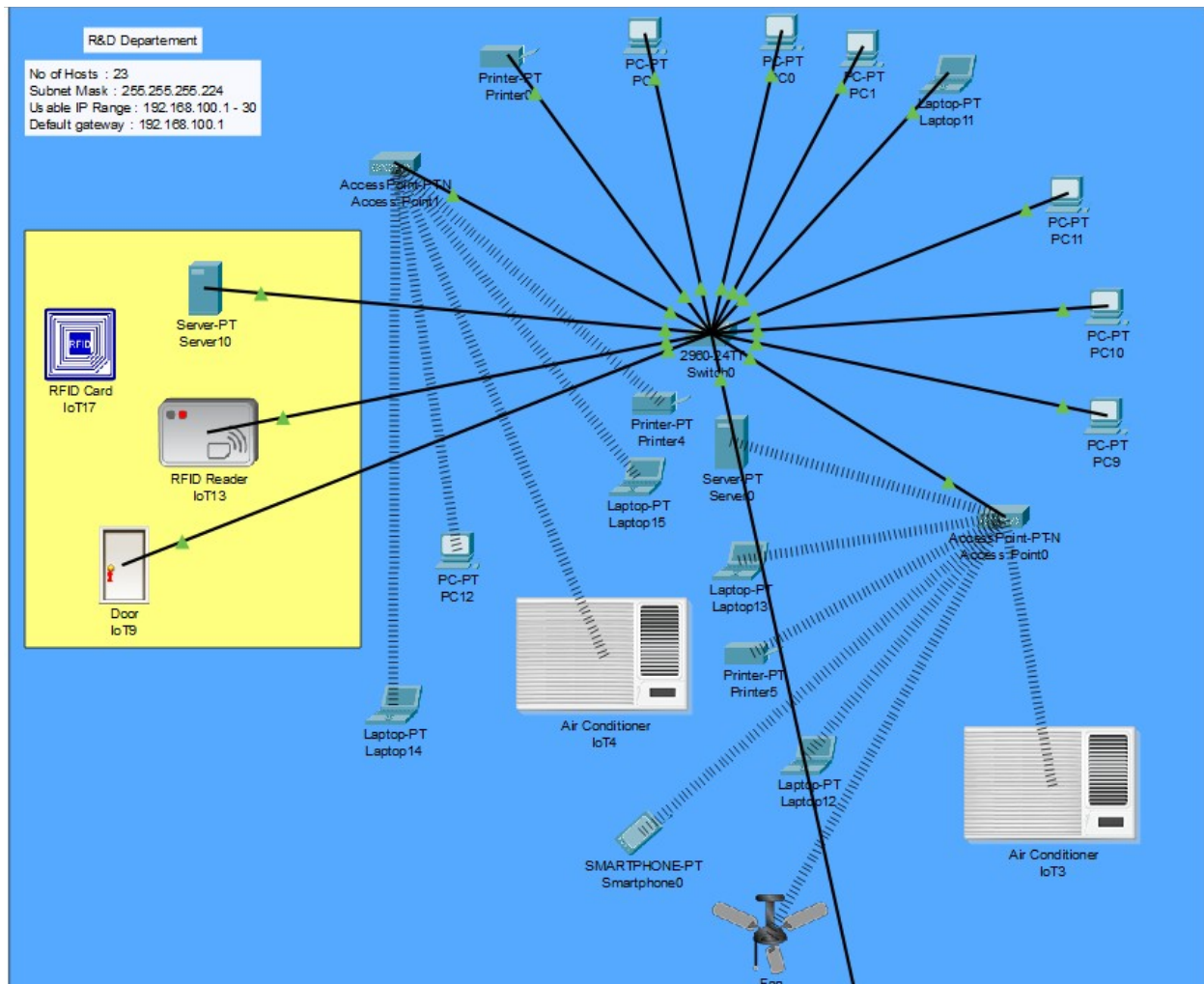
In the modern-day world, where networks are built wherever they are needed, it is essential to design the networks effectively when it comes to ensuring the smooth flow of communication, scalability and

security among different departments of an organization. The present report provides a broad-based subnetting plan to a multi departmental enterprise network covering the best utilization of IP address space, connection of devices and up-scale development.

The network itself has been divided into the main departments: Researches and Development (R&D), Marketing and Sales, Technical support, Server Room, with each host having specific host requirements. In order to make the address as efficient as possible, Variable-Length Subnet Masking (VLSM) is used in the report, enabling use of subnet sizes that fit specific requirements and grow with them at the same time wasting as little IP as possible. Also, to facilitate a higher level of security and automate the working process, IoT gadgets (RFID accessibility, automatic doors, and environmental sensors) are deployed in the design.



2.0 R&D Department :



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R&D Departement

No of Hosts : 23

Subnet Mask : 255.255.255.224

Usable IP Range : 192.168.100.1 - 30

Default gateway : 192.168.100.1

The Reason to Use this Subnet Mask:

Host Requirements: 23 hosts are available in R&D department. A 255.255.255.224 (/27) subnet mask supplies:

The complete IP numbers are 32 (with 30 usable sum, because the first one should be allocated to the network ID, and the last to the broadcast).

Adequate Size: This will support the 23 hosts leaving some expansion space (7 additional addresses).

Effective Addressing: The /27 subnet is convenient, and recommended in small to medium size networks as a compromise with address conservation and ease of use. Severe wastage of addresses would occur in larger subnets (e.g. /26), whereas smaller ones (e.g. /28) would not be enough.

Default Gateway: The first available IP (which is 192.168.100.1) has been configured as the gateway, as is usual to have consistency.

Configure from the router :

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.100.1 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
```

So What is the Rationale of This Configuration?

FastEthernet 0/0 interface in the router would be set-up as a default gateway (192.168.100.1) in the R&D subnet.

The subnet mask (255.255.255.224) will provide 30 usable hosts fitting within the R&D department with 23 hosts, which leave provisions, given the expansion possibility.

No shutdown is absolutely important otherwise the interface would be still disabled and devices would not be able to communicate via the gateway.

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Configure from the switch :

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.100.2 255.255.255.224
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to
up

Switch(config-if)#exit
```

Its configuration starts with entering global configuration mode after which the interface number 1 in the switch is found in the VLAN 1 as the default management VLAN along which IP address is assigned. VLAN 1 is usually associated with simple configurations, but in contemporary security best practice management traffic should be segregated into its own VLAN to reduce risk. However, the VLAN 1 is assigned the IP address as 192.168.100.2 with the subnet mask of 255.255.255.224 (/27) that puts it on the same subnet as the gateway of the router (192.168.100.1). Such synchronization will establish smooth communication between the switch and other equipments in the R&D network.

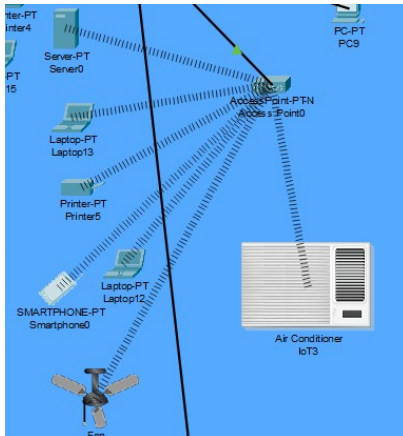
```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip default-gateway 192.168.100.1
Switch(config)#exit
Switch#
```

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These steps will be initiated with the enable command, which will enable us to access the privileged EXEC mode, and then by entering in the global configuration mode, configure terminal. In this case; the administrator would like to set the default gateway to be 192.168.100.1 using option of ip default-gateway 192.168.100.1. This command will represent the interface (192.168.100.1) of the router as the gateway of the switch that will enable it to redirect data to other networks when needed.

The configurations for the end devices (IP – Subnet mask – Default gateway) :

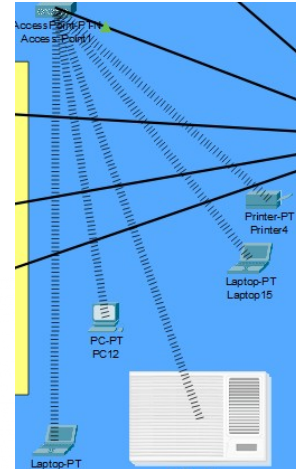
Device	IP Address	Subnet Mask	Default Gateway
1	192.168.100.25	255.255.255.224	192.168.100.1
2	192.168.100.24	255.255.255.224	192.168.100.1
3	192.168.100.23	255.255.255.224	192.168.100.1
4	192.168.100.22	255.255.255.224	192.168.100.1
5	192.168.100.21	255.255.255.224	192.168.100.1
6	192.168.100.20	255.255.255.224	192.168.100.1



IP address : 192.168.100.3 - 9

Subnet Mask :
255.255.255.240

Default gateway : 192.168.100.1

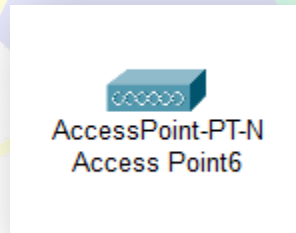


IP address : 1

Subnet Mask

Default gateway

How to create access point :



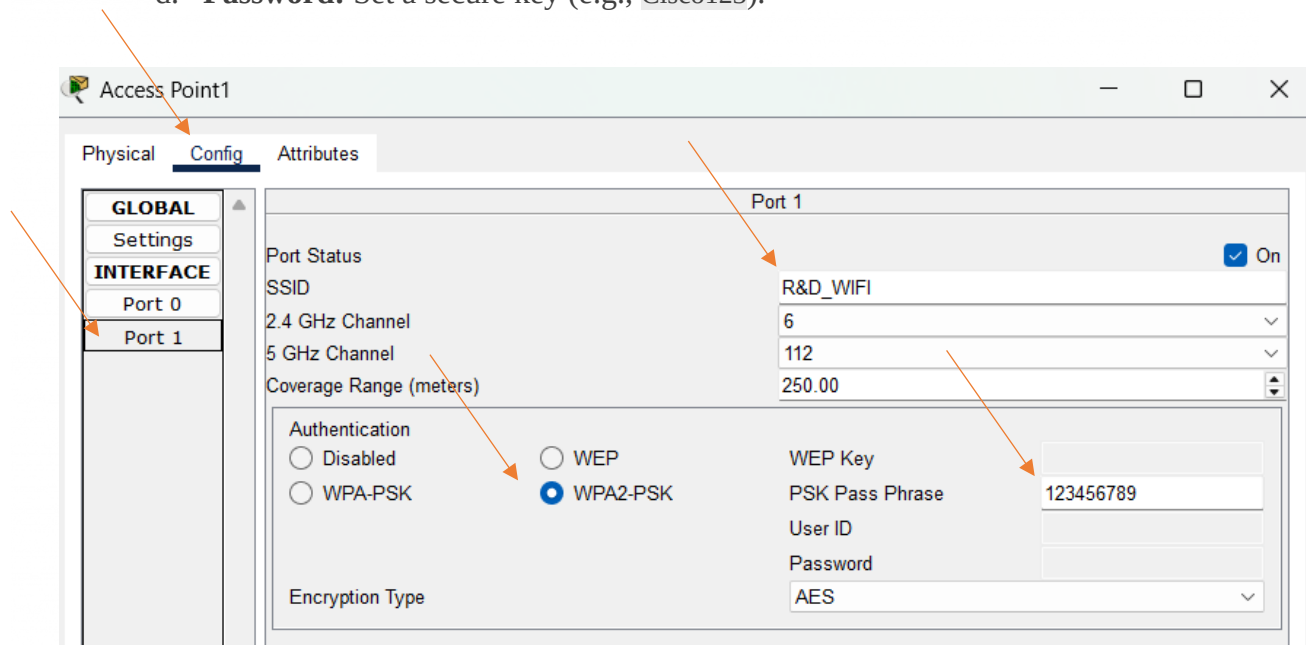
1- add a Wireless Router

Click Wireless, Select > go to Devices Wireless Select e.g. Wireless Router0 (Wireless-PT).

Drag it to the working area.

2- Configure Wireless Settings

- a. Go to **config** then select **port 1**.
- b. **SSID:** Set a name (e.g., MyWiFi).
- c. **Authentication:** WPA2-PSK (recommended).
- d. **Password:** Set a secure key (e.g., Cisco123).



How to connect with wireless :

- 1- Open end device then go to desktop

- 2- Then click on PC wireless



- 3- Search and select the wifi

Link Information

Connect

Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
R&D_WIFI_2	1	63%
Marketing & Sales	1	63%
Technical_support...	1	63%
Default	1	63%

Refresh

Connect

Site Information

Wireless Mode

Infrastructure

Network Type

Mixed B/G/N

Radio Band

Auto

Security

WPA2-PSK

MAC Address

0050.0F39.7676

WPA2-Personal Needed for Connection

This wireless network has WPA2-Personal enabled. To connect to this network, enter the required passphrase in the appropriate field below. Then click the **Connect** button.

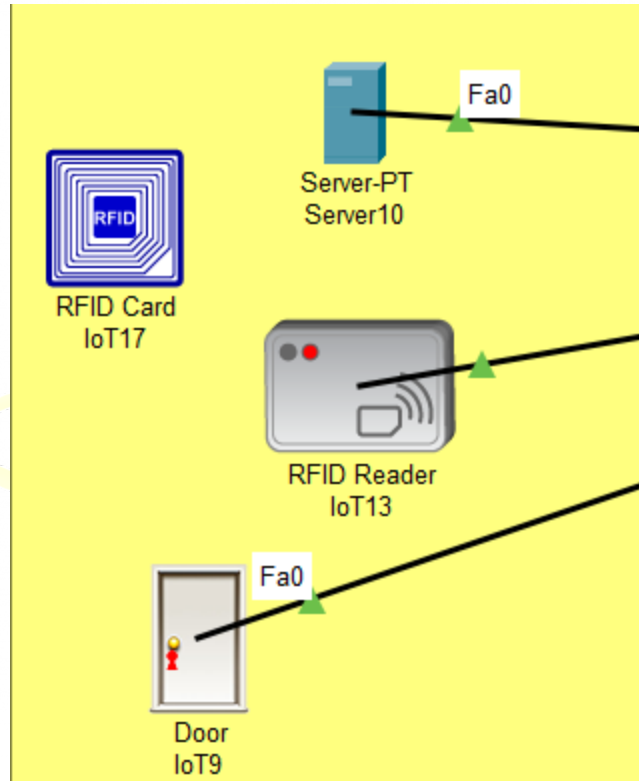
Security

WPA2-Personal

Please select the wireless security method used by your existing wireless network.

Pre-shared Key

Please enter a Pre-shared Key that is 8 to 63 characters in length.



RFID Card (IoT17) and RFID Reader (IoT13)

Creates an RFID based access control system where the value of the card will be scanned by the reader.

Pros: Allows the use of contact-less buildings or restricted areas entrance systems.

Server-PT (Server10)

Performs the role of the central management system, so it is probably processing data of the RFID system and IoT devices.

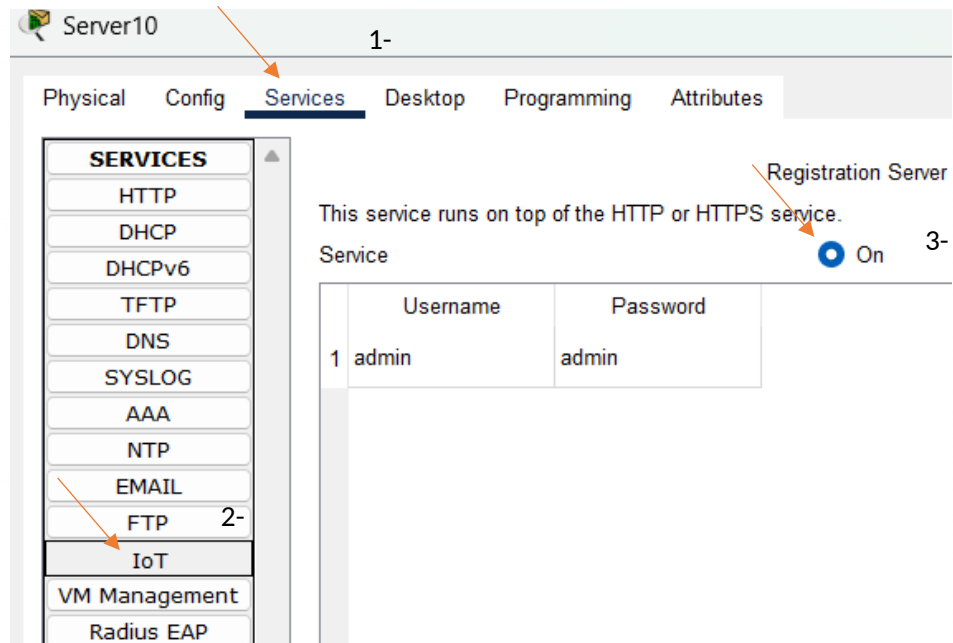
Pro: Stores, processes, and controls policies of data access into a centralized place enabling efficient management.

Door (IoT9)

Stands for an automated door operated with the help of the RFID system.

Pro: It is more secure since only verified individuals with the appropriate security clearance (RFID validation) can access the possibly restricted area.

1- configure form Server-PT (Server10) :



2- Configure from the door (IoT9) and RFID Reader (IoT13) :

IoT Server

☐ None

☐ Home Gateway

☒ Remote Server

Server Address

User Name

Password

IoT Server

☐ None

☐ Home Gateway

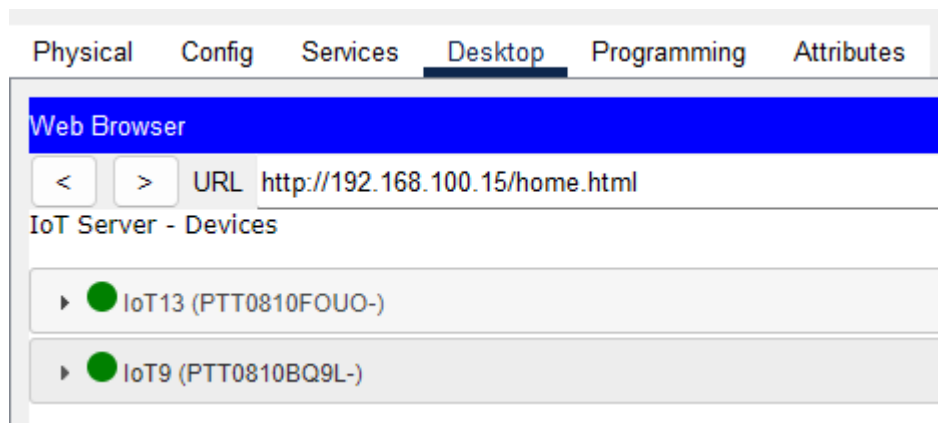
☒ Remote Server

Server Address

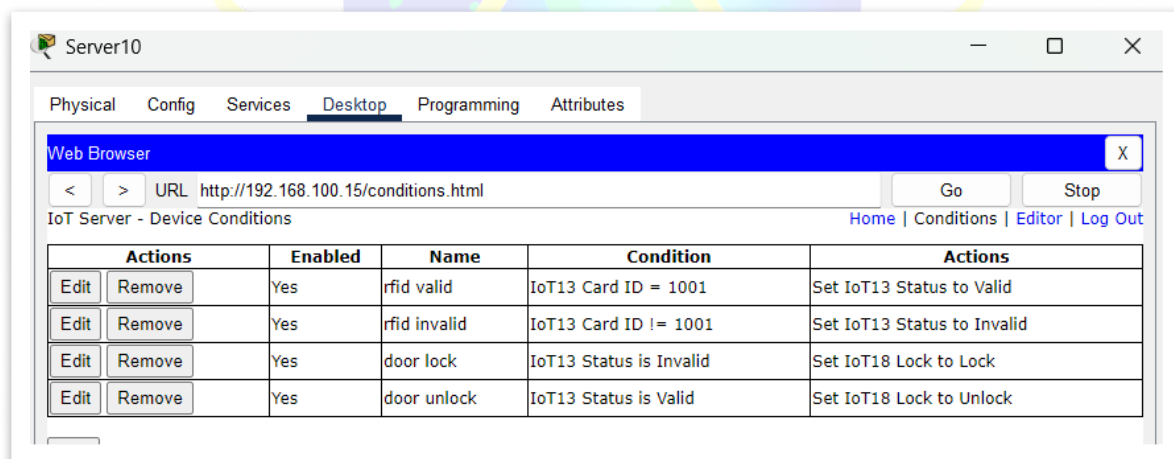
User Name

Password

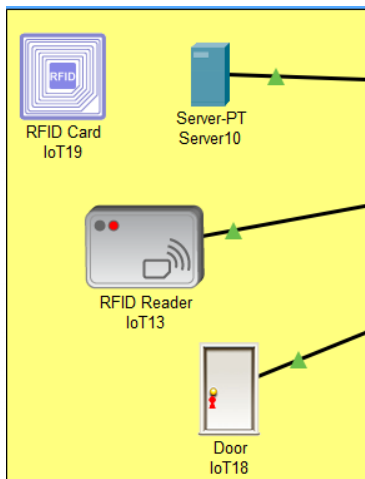
3. Check the end devices are connected to the server :



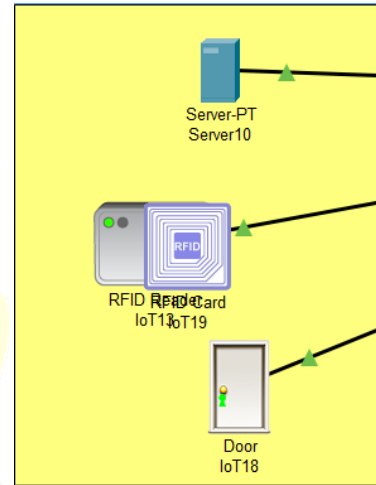
4- Add the conditions



5- Check the conditions if it works :



-- Before --

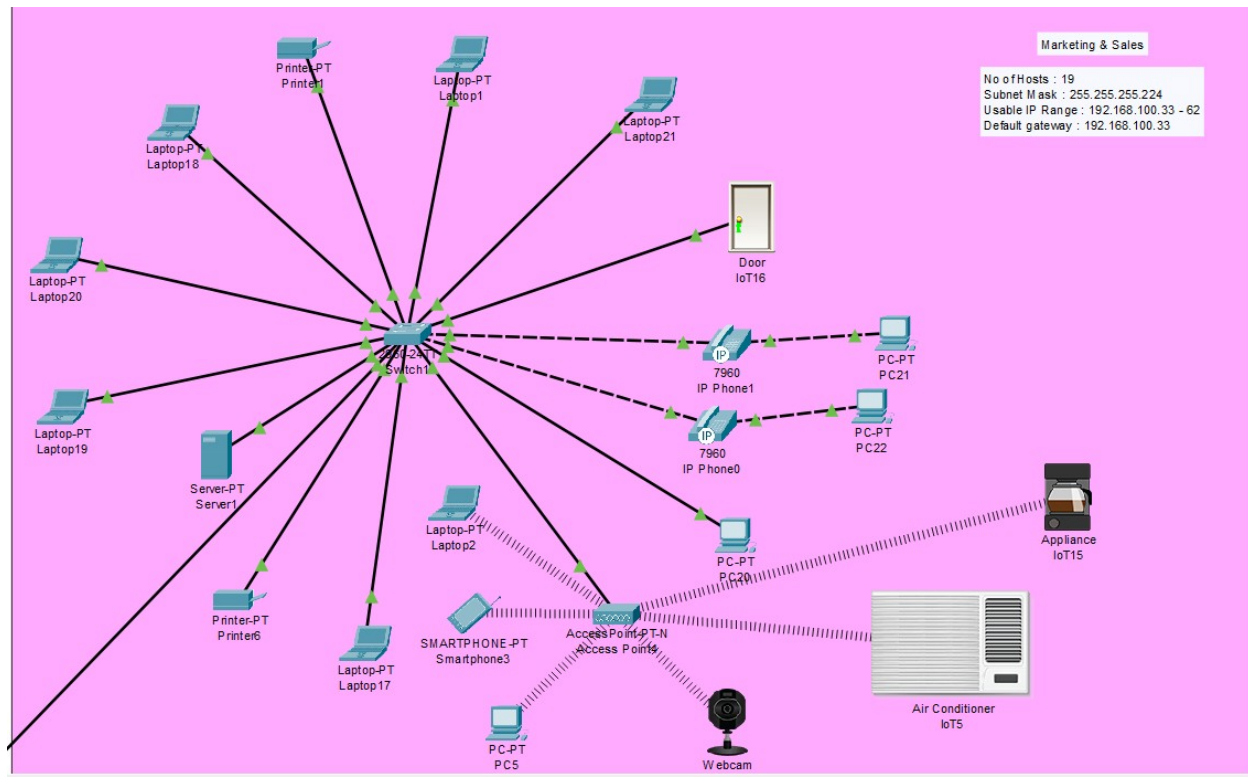


-- After --

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3.0 Marketing & Sales Department:

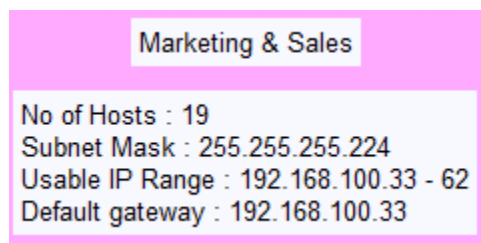


Subnet Design: Our department uses the helper subnet 192.168.100.32/27 (mask: 255.255.255.224), and offers 30 useful IP addresses (33 62) to host 19 systems and has a remaining capability of expansion.

The routing interface is default gateway: 192.168.100.33 through which it can communicate to other network segments.

Connected Devices: Between workstations (laptops, PCs), servers, smartphones, and IoT (e.g., "All Conditioner IoT5") are used in the department--this is nearly an ideal combination of connectivity needs in the department.

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- 1- Scalability: 11 free assignable addresses to support extra devices without reconfiguration

- 2- Isolation: Isolated subnet More security and traffic control offense
- 3- Uniformity: Inconsistency /27 subnetting of departments makes administration easy
- 4- Optimization: No cognizant addresses forfeited (relative to the utilization of a bigger /26 subnet)

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet7/0
Router(config-if)#ip address 192.168.100.33 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config) #
```

The example cited in this configuration snippet is establishment of a GigabitEthernet interface on a Cisco router, which is a primary activity during implementation of networks. The commands used here are the commands which assign an IP to the interface and displays the interface to be in operation so that it is able to engage in network communication.

Network Segmentation: The /27 subnet is a good physical division of the network as it accepts medium-sized network segments such as server farms or departmental LANs.

Speedy Connectivity: The GigabitEthernet guarantees quick transmission of data, appropriate in high bandwidth activities.

Operation Preparedness - The no shutdown should be entered vital - Interfaces are defaulted to administratively down and will not operate until activated.

Configure the Switch (VLAN 1 + Default Gateway)

```
Marketing&sales>en
Marketing&sales#confi
Marketing&sales#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Marketing&sales(config)#interface v
Marketing&sales(config)#interface vlan 1
Marketing&sales(config-if)#ip address 192.168.100.34 255.255.255.224
Marketing&sales(config-if)#no sh
Marketing&sales(config-if)#no shutdown
Marketing&sales(config-if)#exit
Marketing&sales(config)#
```

Configuration of VLAN Interface VLAN

VLAN one is set up with an IP of 192.168.100.34/27

This makes it on the same subnet as earlier devices (192.168.100.32/27 network)

Subnet Mask 255.255.255.224 gives 30 Available host addresses

Set the default gateway (router's IP)

```
Marketing&sales#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Marketing&sales(config)#interface vlan1
Marketing&sales(config-if)#ip default-gateway 192.168.100.33
Marketing&sales(config)#exit
Marketing&sales#
%SYS-5-CONFIG_I: Configured from console by console
```

The configurations for the end devices (IP – Subnet mask – Default gateway) :

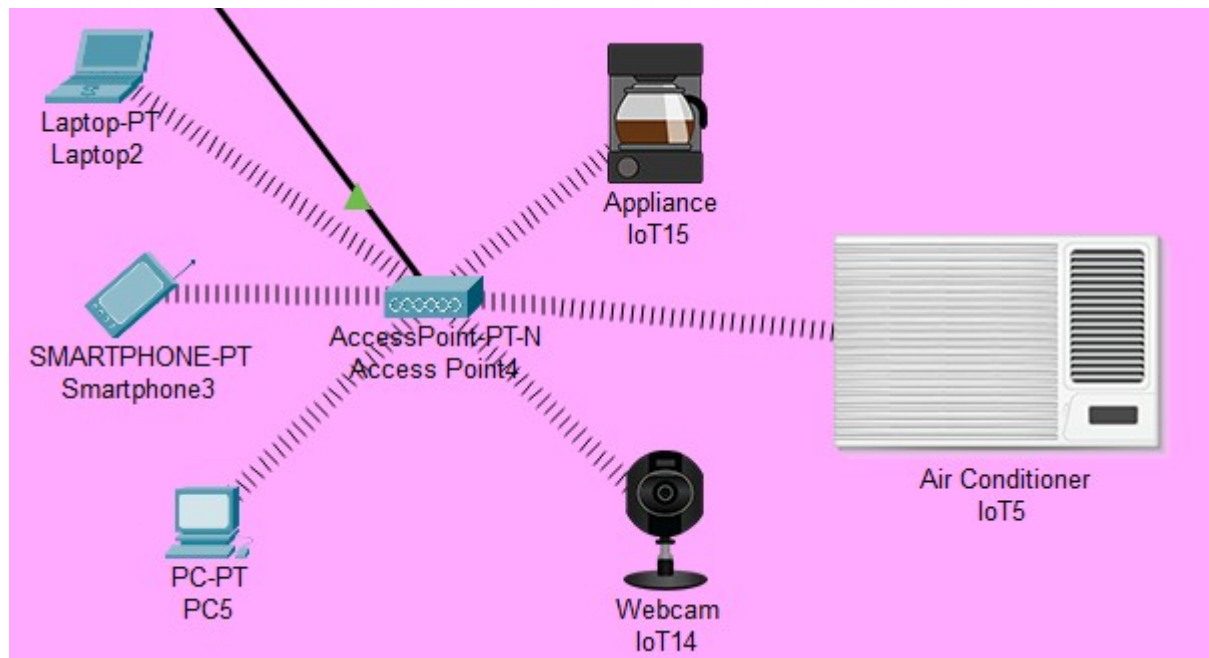
IP Configuration		☒ Static
☐ DHCP		
IPv4 Address	192.168.100.48	192.168.100.49
Subnet Mask	255.255.255.224	255.255.255.224
		192.168.100.33
		0.0.0.0

☒ Static	☒ Static	☒ Static
192.168.100.50	192.168.100.52	192.168.100.53
255.255.255.224	255.255.255.224	255.255.255.224
192.168.100.33	192.168.100.33	192.168.100.33
0.0.0.0	0.0.0.0	0.0.0.0

IP Configuration		☒ Static
☐ DHCP		
IPv4 Address	192.168.100.51	192.168.100.35
Subnet Mask	255.255.255.224	255.255.255.224
		192.168.100.33
		0.0.0.0
Default Gateway		192.168.100.33

☒ Static	☒ Static
192.168.100.36	192.168.100.38
255.255.255.224	255.255.255.224
192.168.100.33	192.168.100.33
0.0.0.0	0.0.0.0

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.100.37
Subnet Mask	255.255.255.224



IP address : 192.168.100.39 – 44

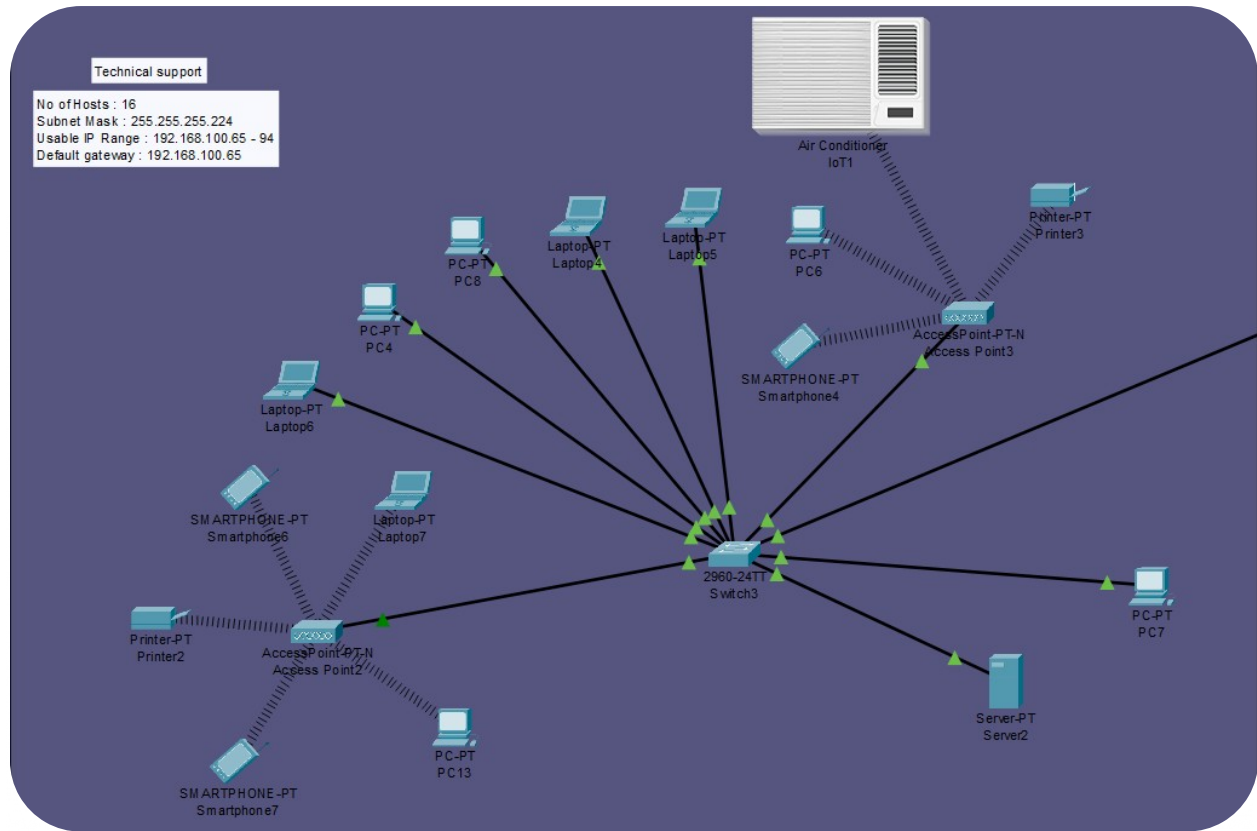
Subnet mask : 255.255.255.224

Default – gateway : 192.168.100.33

☒ Static ☐ DHCP	
192.168.100.46	192.168.100.47
255.255.255.224	255.255.255.224
192.168.100.33	192.168.100.33
0.0.0.0	0.0.0.0

IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	192.168.100.48
Subnet Mask	255.255.255.224

4.0 Technical Support Department:

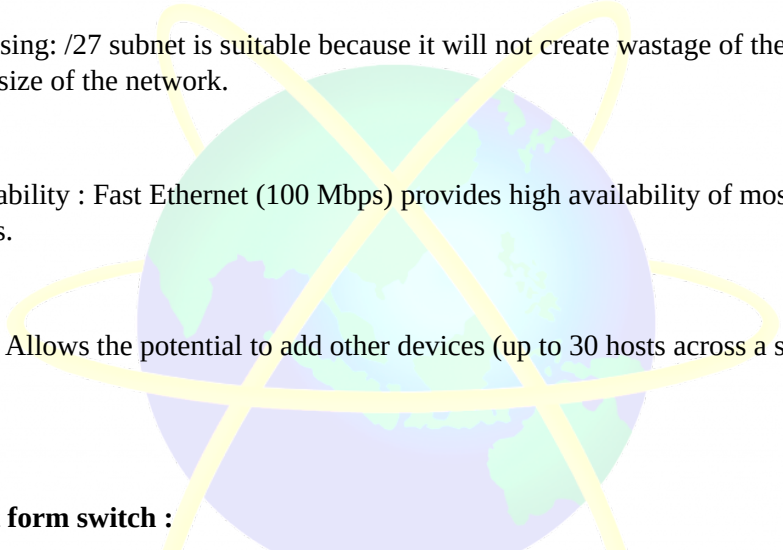


This design describes the backbone network of a Technical Support division, including a well-planned IP addressing system and attached devices that support the operations of the employees as well as IoT-enhanced environmental controls.

- **IP Allocation:**

- o Usable Range: 192.168.100.65 - 94
- o Gateway: 192.168.100.65 (router interface)
- o Reserved Addresses: 192.168.100.64 (network), 192.168.100.95 (broadcast)

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 1/0
Router(config-if)#ip address 192.168.100.65 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
```

- 
- 1- Best addressing: /27 subnet is suitable because it will not create wastage of the IP addressing against the size of the network.
 - 2- High Availability : Fast Ethernet (100 Mbps) provides high availability of most LAN applications.
 - 3- Scalability: Allows the potential to add other devices (up to 30 hosts across a subnet).

Technical Support form switch :

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip address 192.168.100.66 255.255.255.224
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Meaning and Usefulness :

Remote Management: The IP provides SSH/telnet access to switch.

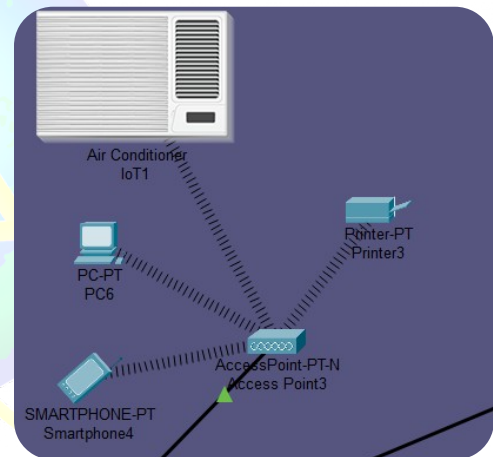
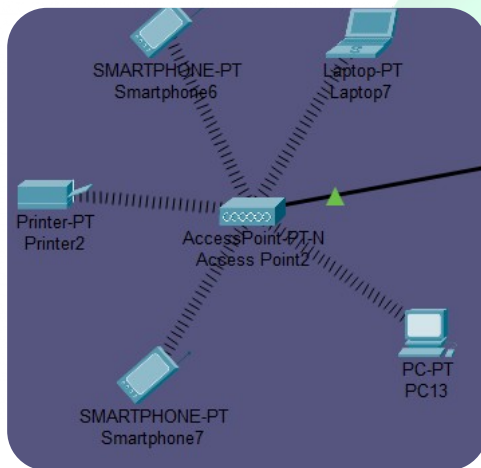
Inter-VLAN Routing: In case the switch is a Layer 3 capable switch, then this SVI can forward traffic between VLANs.

Subnet Consistency: The subnet of the device is consistent with that of other devices (i.e. router of 192.168.100.65).

Set the default gateway (router's IP) :

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan1
Switch(config-if)#ip default-gateway 192.168.100.65
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

The configurations for the end devices (IP – Subnet mask – Default gateway) :



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IP address : 192.168.100.78 – 82

Subnet mask : 255.255.255.224

Default – gateway : 192.168.100.65

IP address : 192.168.100.83 – 90

Subnet mask : 255.255.255.224

Default – gateway : 192.168.100.65

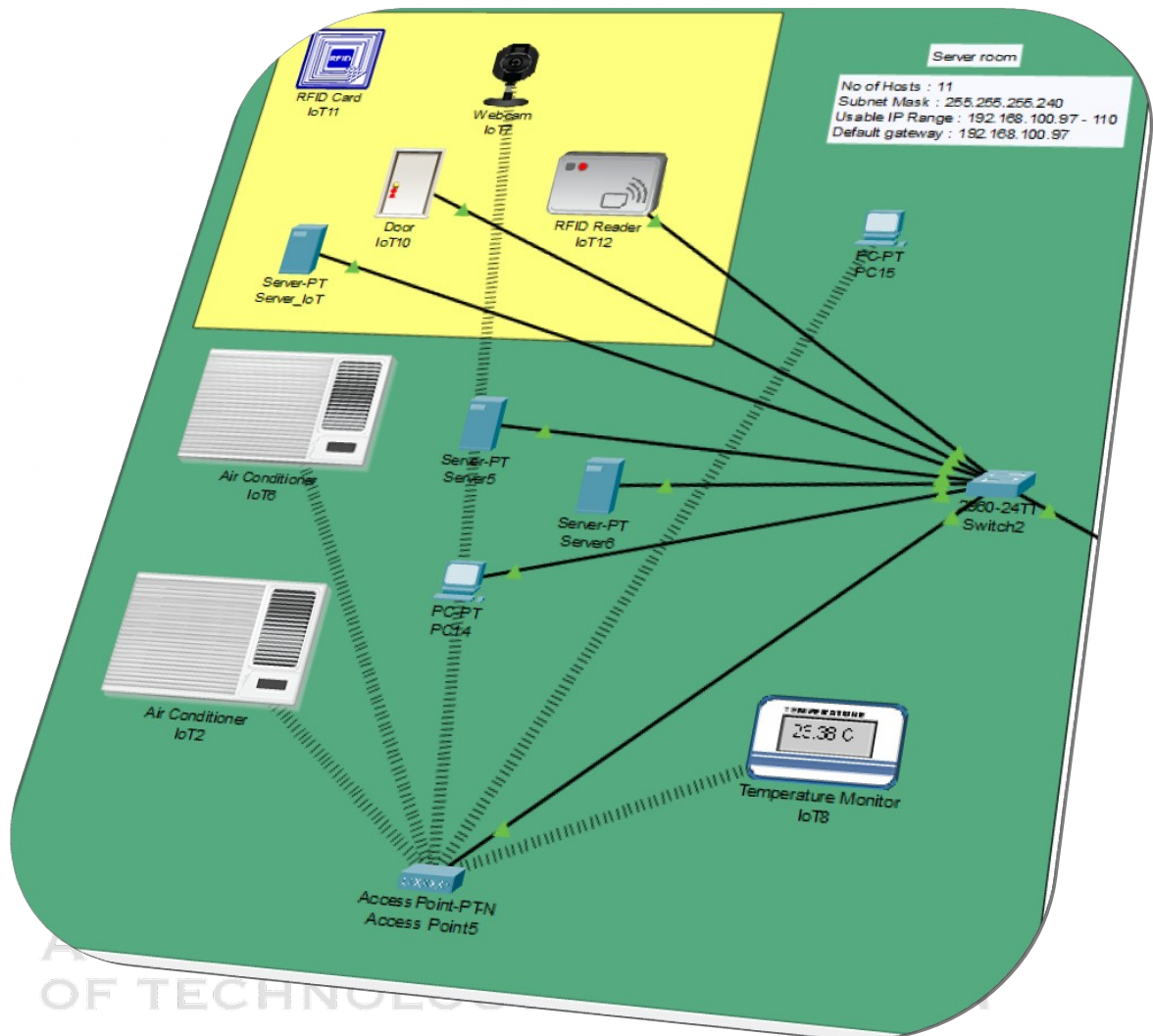
☒ Static	☒ Static	☒ Static
192.168.100.73	192.168.100.74	192.168.100.75
255.255.255.224	255.255.255.224	255.255.255.224
192.168.100.65	192.168.100.65	192.168.100.65
0.0.0.0	0.0.0.0	0.0.0.0

☒ Static	☒ Static	☒ Static
192.168.100.76	192.168.100.77	192.168.100.67
255.255.255.224	255.255.255.224	255.255.255.224
192.168.100.65	192.168.100.65	192.168.100.65
0.0.0.0	0.0.0.0	0.0.0.0

☒ Static
192.168.100.68
255.255.255.224
192.168.100.65
0.0.0.0

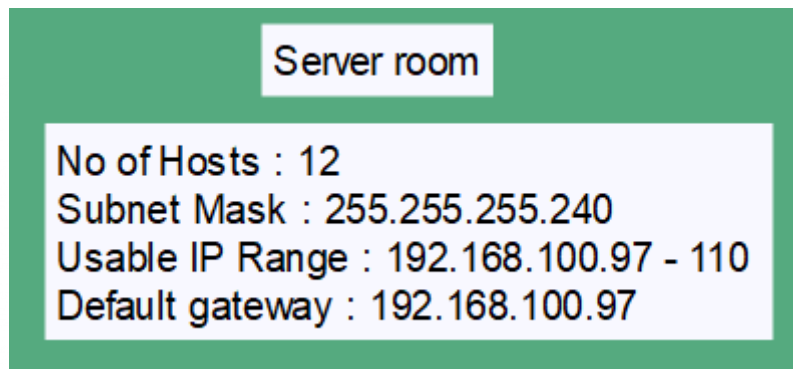
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5.0 Server Room Department :



The design introduces a well-balance yet modernistic server room environment with a strategic integration of the classical network infrastructure and IoT-connected devices to develop a safe and regulated climate-controlled monitoring and carefully controlled technical area. All these are displayed in

the network architecture where the new trend of server rooms can provide the ideal operating situations with the provision of security to the hardware physically.



1. Addressing Scheme

- **Subnet:** 192.168.100.96/28
- **Subnet Mask:** 255.255.255.240
- **Total Addresses:** 16 (14 usable)
- **Current Utilization:** 12 hosts (86% capacity)
- **IP Allocation:**
 - o Network Address: 192.168.100.96
 - o Usable Host Range: 192.168.100.97-110
 - o Gateway: 192.168.100.97 (router interface)
 - o Broadcast: 192.168.100.111

Server Room from the router

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 6/0
Router(config-if)#ip address 192.168.100.97
% Incomplete command.
Router(config-if)#ip address 192.168.100.97 255.255.255.240
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Configuration Breakdown

1. Initial Setup :

- User goes to user EXEC (Router>) mode to privileged EXEC (Router#) mode
- Gets into the global configuration (config t) mode

2. Interface Selection :

- Chooses FastEthernet6/0 interface
- Trenching is the proper selection of a server room connection (wired reliability)

3. Distribution of IPs Process :

- First try (ip address 192.168.100.97) ends in error "% Incomplete command"
- Fixed by introducing subnet mask (255.255.255.240)
- Final issue: ip address 192.168.100.97 255.255.255.240

Server Room form switch :

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.100.98 255.255.255.240
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

1. Interface Configuration :

Switch virtual interface (SVI) VLAN 1:

- IP: 192.168.100.98
- Subnet Mask 255.255.255.240 (/28)
- Second Useable IP in 192.168.100.96/ 28 Subnet
- Router gateway (192.168.100.97) router adjacency

2. Operational Status

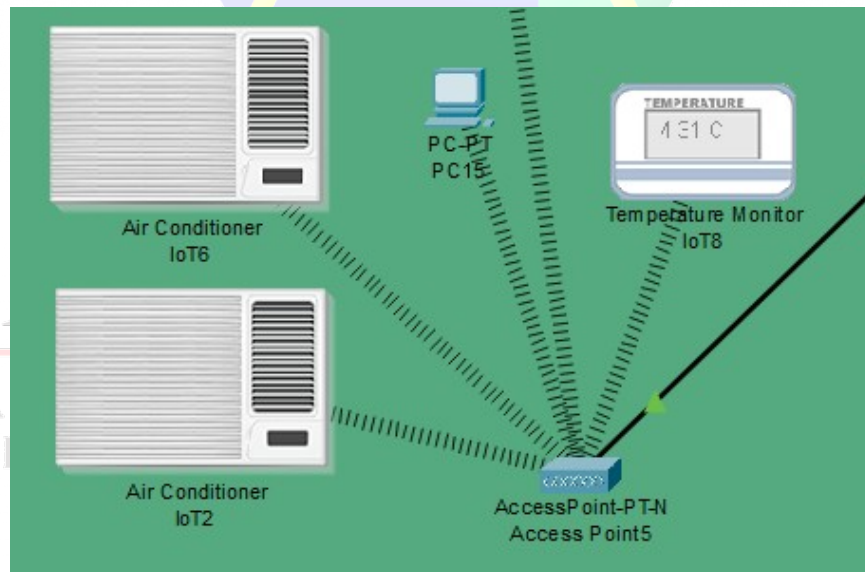
Interface enabled by non shutdown

- Automatic acknowledgement of status of link/line protocol changes
- There are no error messages that show successful implementation

Set the default gateway (router's IP) :

```
Switch(config)#interface vlan1
Switch(config-if)#ip default-gateway 192.168.100.97
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

The configurations for the end devices (IP – Subnet mask – Default gateway) :

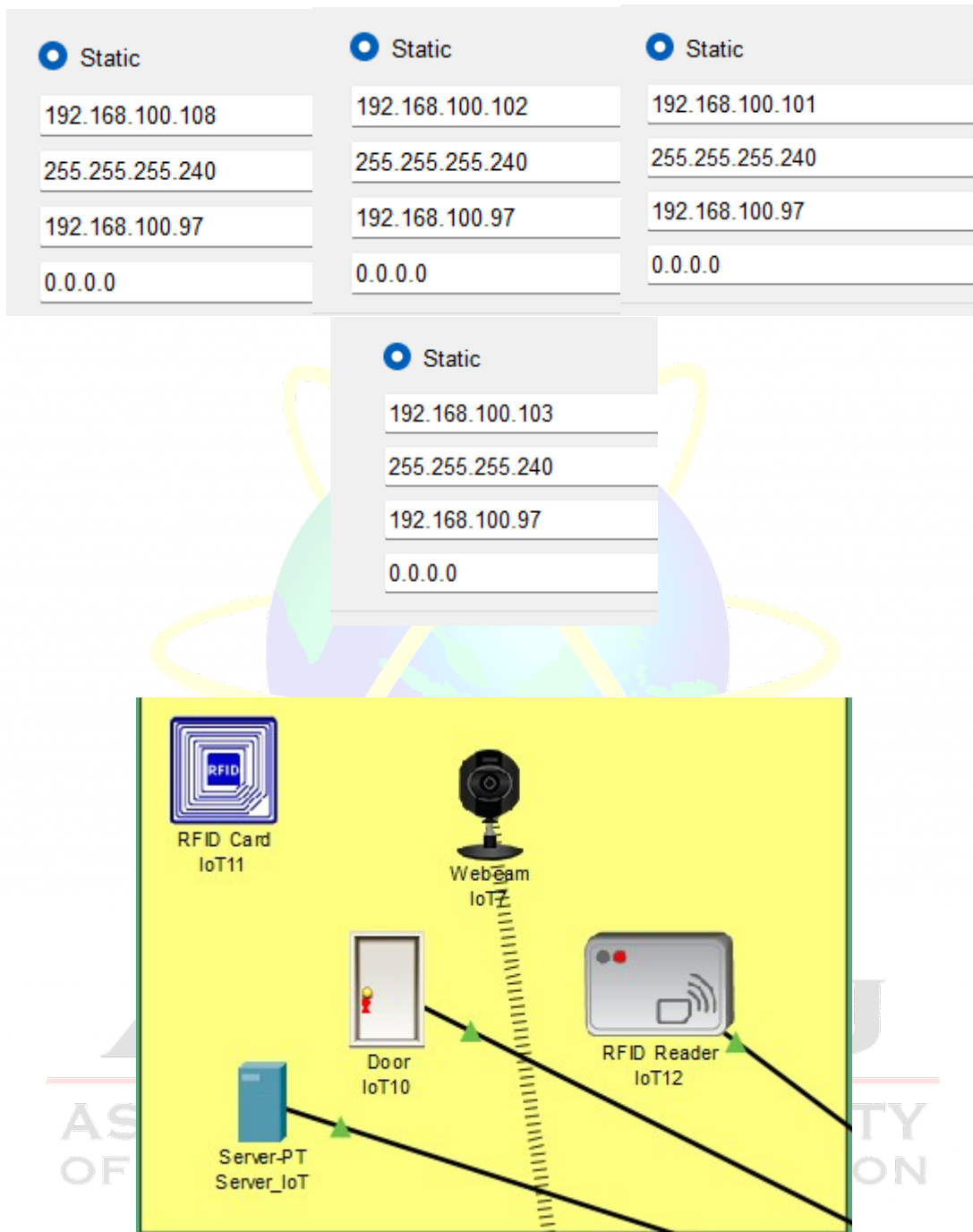


IP address : 192.168.100.104 – 106

And .108

Subnet mask : 255.255.255.224

Default – gateway : 192.168.100.65



Apparatus Stock & Usage

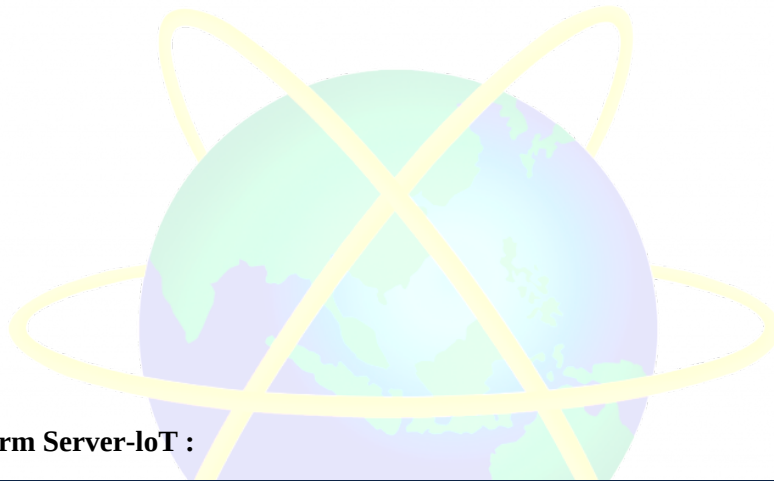
1. Access Control System

- RFID Reader (IoT12): verifies identities
- RFID Card (IoT11): Workforce authorizer

- Automated Door (IoT10): Automated electromechanical entrance point

2. Surveillance System

- Webcam (IoTZ): Camera (Imaging)
- Server Integration
- Server_IoT: Management central of security installation



1. configure form Server-IoT :

The screenshot shows the 'Server_IoT' configuration window. The 'Services' tab is active, displaying a list of services on the left and configuration details on the right. The 'IoT' service is selected in the list. The right panel shows the 'Registration Server' status as 'On' and a table with one user entry.

SERVICES	
HTTP	
DHCP	
DHCPv6	
TFTP	
DNS	
SYSLOG	
AAA	
NTP	
EMAIL	
FTP	
IoT	
VM Management	
Radius EAP	

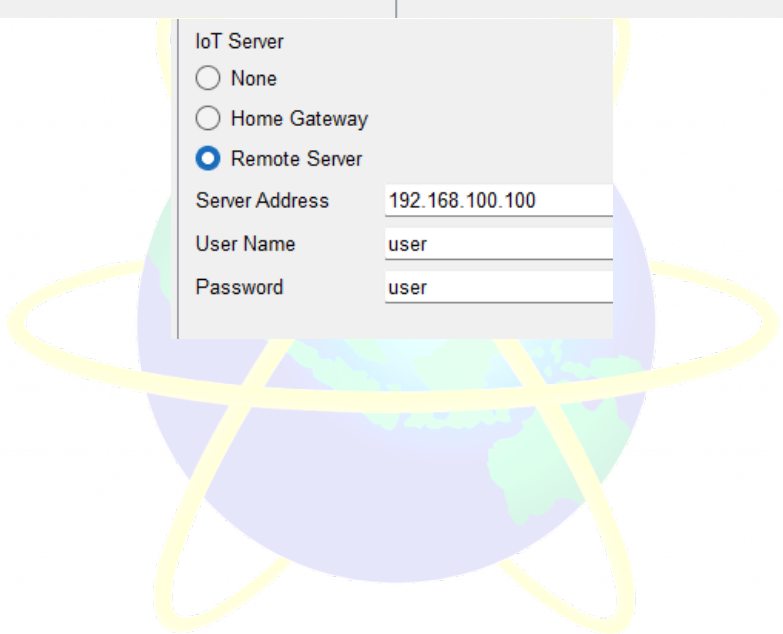
Registration Server
This service runs on top of the HTTP or HTTPS service.

Service ☒ On

	Username	Password
1	user	user

2. Configure from the door , RFID Reader and the webcam :

IoT Server		IoT Server	
☐ None		☐ None	
☐ Home Gateway		☐ Home Gateway	
☒ Remote Server		☒ Remote Server	
Server Address	192.168.100.100	Server Address	192.168.100.100
User Name	user	User Name	user
Password	user	Password	user



IoT Server	
☐ None	
☐ Home Gateway	
☒ Remote Server	
Server Address	192.168.100.100
User Name	user
Password	user

6- Check the end devices are connected to the server :

Server_IoT

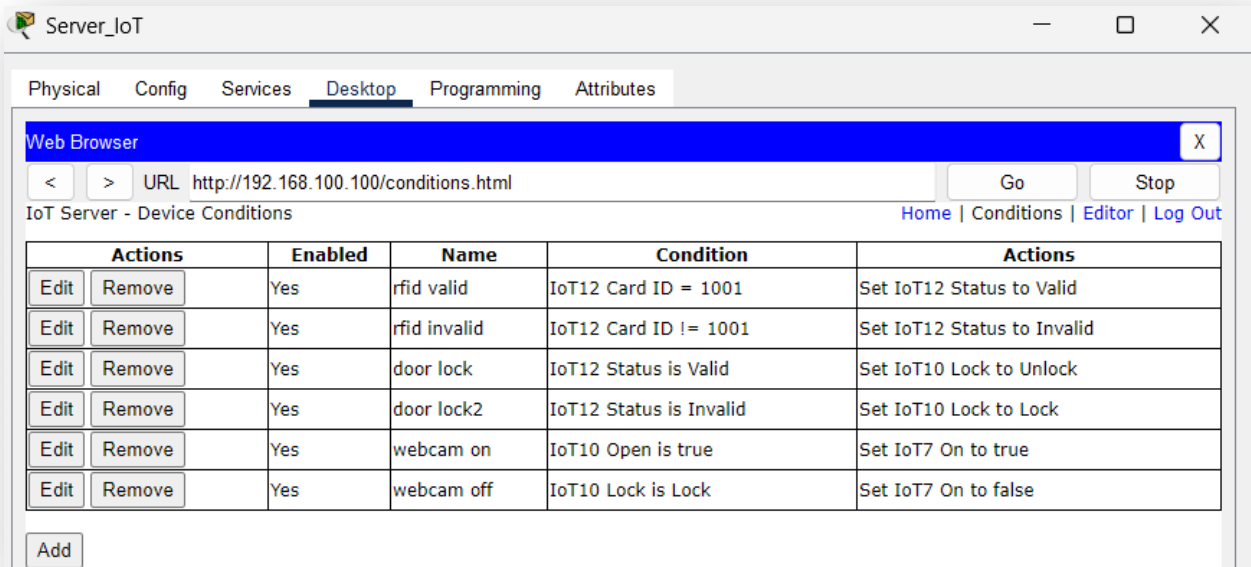
Physical Config Services **Desktop** Programming Attributes

IoT Monitor X

IoT Server - Devices Home | Conditions | Editor | Log Out

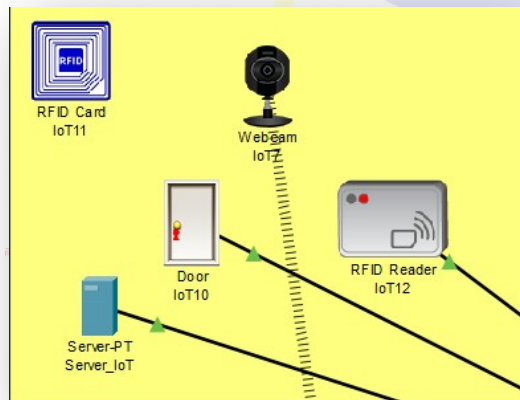
▶ ● IoT12 (PTT08102LB7-)	RFID Reader
▶ ● IoT10 (PTT081015C0-)	Door
▶ ● IoT7 (PTT0810JO82-)	Webcam

7- Add the conditions

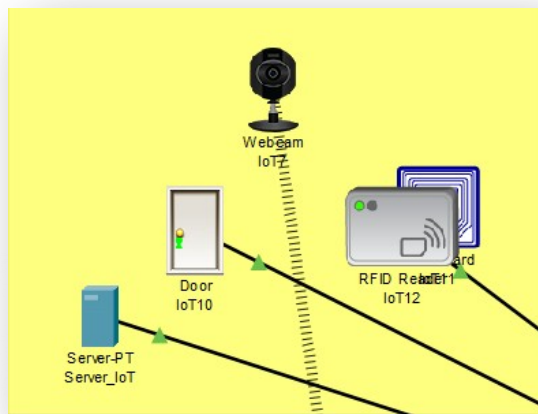


Note : the webcam will turn on only when the door open .

8- Check the conditions if it works :



-- Before --



-- After --

6.0 Subnetting Scheme Table :

Which Subnetting Strategy we have to select ??

We have two options :

1. **Fixed-Length Subnet Mask (FLSM):** All subnets same size
2. **Variable-Length Subnet Mask (VLSM):** Subnets can be different sizes

Why I chose VLSM:

- More efficient use of IP space
- Allows better accommodation of different size requirements
- Leaves more room for future expansion

Subnet Name	No of Hosts	Block Size (2^n)	Subnet Mask	Network Address	Broadcast Address	Usable IP Range
R&D	23	32 (2^5)	255.255.255.224	192.168.100.0	192.168.100.31	192.168.100.1 - 192.168.100.30
Marketing & Sales	19	32	255.255.255.224	192.168.100.32	192.168.100.63	192.168.100.33 -

		(2 ⁵)				192.168.100.62
Technical Support	16	32 (2 ⁵)	255.255.255.224	192.168.100.64	192.168.100.95	192.168.100.65 - 192.168.100.94
Server Room	11	16 (2 ⁴)	255.255.255.240	192.168.100.96	192.168.100.111	192.168.100.97 - 192.168.100.110
Router Link 1	2	4 (2 ²)	255.255.255.252	192.168.100.112	192.168.100.115	192.168.100.113 - 192.168.100.114
Future Expansion	-	32 (2 ⁵)	255.255.255.224	192.168.100.120	192.168.100.151	192.168.100.121 - 192.168.100.150

1. R&D (20 hosts):

- o $2^5 = 32 \rightarrow 32 - 2 = 30$ usable IPs (meets 20 requirement with 10 extra)
- o Next size down ($2^4 = 16$) would only give 14 usable IPs (too small)

2. Marketing (20 hosts):

- o Same as R&D $\rightarrow$ needs /27 (32 IP block)

3. Technical Support (17 hosts):

- o Could use $2^5 = 32$ (30 usable) to match others and allow growth
- o Alternative: $2^4 = 16$ (14 usable) would barely meet current needs but leave no room for growth.

4. Server Room (11 hosts):

- o $2^4 = 16 \rightarrow 16 - 2 = 14$ usable IPs (meets 11 with 3 extra)
- o Next size down ($2^3 = 8$) would only give 6 usable IPs (too small)

5. Router Links:

- o Each link needs exactly 2 usable IPs (one for each router interface)
- o $2^2 = 4 \rightarrow 4 - 2 = 2$ usable IPs (perfect fit)

6- Future Expansion

- Left one /27 block (32 IPs):
 - o Network: 192.168.100.120
 - o Broadcast: 192.168.100.151
 - o Could use $2^5 = 32$ (30 usable).

7.0 Testing & commissioning :

7.1 D&R Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	Laptop11	ICMP		0.000	N	0	(edit)	
	Successful	Lapto...	Laptop15	ICMP		0.000	N	1	(edit)	
	Successful	PC11	Laptop19	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Lapto...	PC15	ICMP		0.000	N	3	(edit)	
	Successful	PC12	Router2	ICMP		0.000	N	4	(edit)	
	Successful	Printer5	Laptop5	ICMP		0.000	N	5	(edit)	

```
C:\>ping 192.168.100.100

Pinging 192.168.100.100 with 32 bytes of data:

Reply from 192.168.100.100: bytes=32 time=10ms TTL=127
Reply from 192.168.100.100: bytes=32 time=22ms TTL=127
Reply from 192.168.100.100: bytes=32 time=9ms TTL=127
Reply from 192.168.100.100: bytes=32 time=40ms TTL=127

Ping statistics for 192.168.100.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 40ms, Average = 20ms

C:\>
```

```
C:\>ping 192.168.100.73

Pinging 192.168.100.73 with 32 bytes of data:

Reply from 192.168.100.73: bytes=32 time=35ms TTL=127
Reply from 192.168.100.73: bytes=32 time=52ms TTL=127
Reply from 192.168.100.73: bytes=32 time=24ms TTL=127
Reply from 192.168.100.73: bytes=32 time=6ms TTL=127

Ping statistics for 192.168.100.73:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 52ms, Average = 29ms

C:\>
```

7.2 Marketing & Sales Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Lapto...	PC9	ICMP		0.000	N	0	(edit)	
	Successful	Lapto...	Server5	ICMP		0.000	N	1	(edit)	
	Successful	PC21	Laptop6	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Lapto...	Laptop20	ICMP		0.000	N	0	(edit)	
	Successful	PC20	Switch1	ICMP		0.000	N	1	(edit)	
	Successful	IoT16	Router2	ICMP		0.000	N	2	(edit)	

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.36

Pinging 192.168.100.36 with 32 bytes of data:

Reply from 192.168.100.36: bytes=32 time<1ms TTL=128
Reply from 192.168.100.36: bytes=32 time=1ms TTL=128
Reply from 192.168.100.36: bytes=32 time<1ms TTL=128
Reply from 192.168.100.36: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

```

C:\>ping 192.168.100.76

Pinging 192.168.100.76 with 32 bytes of data:

Reply from 192.168.100.76: bytes=32 time<1ms TTL=127
Reply from 192.168.100.76: bytes=32 time=7ms TTL=127
Reply from 192.168.100.76: bytes=32 time<1ms TTL=127
Reply from 192.168.100.76: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.100.76:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>

```

7.3 Technical Support Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Server2	Laptop4	ICMP		0.000	N	0	(edit)	
	Successful	PC7	PC6	ICMP		0.000	N	1	(edit)	
	Successful	Printer2	Switch3	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC8	PC14	ICMP		0.000	N	0	(edit)	
	Successful	Laptop6	IoT2	ICMP		0.000	N	1	(edit)	
	Successful	IoT1	IoT4	ICMP		0.000	N	2	(edit)	

```
C:\>ping 192.168.100.99

Pinging 192.168.100.99 with 32 bytes of data:

Reply from 192.168.100.99: bytes=32 time<1ms TTL=254
Reply from 192.168.100.99: bytes=32 time<1ms TTL=254
Reply from 192.168.100.99: bytes=32 time=1ms TTL=254
Reply from 192.168.100.99: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.100.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
C:\>ping 192.168.100.40

Pinging 192.168.100.40 with 32 bytes of data:

Reply from 192.168.100.40: bytes=32 time=5ms TTL=127
Reply from 192.168.100.40: bytes=32 time=7ms TTL=127
Reply from 192.168.100.40: bytes=32 time=4ms TTL=127
Reply from 192.168.100.40: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.100.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 6ms

C:\>
```

7.4 Server Room Department :

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC15	Server6	ICMP		0.000	N	0	(edit)	
	Successful	Serve...	IoT7	ICMP		0.000	N	1	(edit)	
	Successful	IoT12	PC14	ICMP		0.000	N	2	(edit)	

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Serve...	PC12	ICMP		0.000	N	2	(edit)	
	Successful	IoT7	PC11	ICMP		0.000	N	3	(edit)	
	Successful	IoT8	Laptop21	ICMP		0.000	N	4	(edit)	

```
C:\>ping 192.168.100.6

Pinging 192.168.100.6 with 32 bytes of data:

Reply from 192.168.100.6: bytes=32 time=33ms TTL=127
Reply from 192.168.100.6: bytes=32 time=6ms TTL=127
Reply from 192.168.100.6: bytes=32 time=18ms TTL=127
Reply from 192.168.100.6: bytes=32 time=18ms TTL=127

Ping statistics for 192.168.100.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 33ms, Average = 18ms

C:\>
```

```
C:\>ping 192.168.100.65

Pinging 192.168.100.65 with 32 bytes of data:

Reply from 192.168.100.65: bytes=32 time<1ms TTL=255
Reply from 192.168.100.65: bytes=32 time<1ms TTL=255
Reply from 192.168.100.65: bytes=32 time<1ms TTL=255
Reply from 192.168.100.65: bytes=32 time=8ms TTL=255

Ping statistics for 192.168.100.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>|
```

1.0 Introduction :

The topology of the network used in this project is Simex which handles the activities of SkyNetix, a company that has four branches across the nation, i.e., Kuala Lumpur, Pulau Pinang, Johor Bahru and Kota Kinabalu. The following documentation is specific to Kota Kinabalu branch that is further subdivided into three critical functional segments of Staff Point, Guest Room, and Server Room.

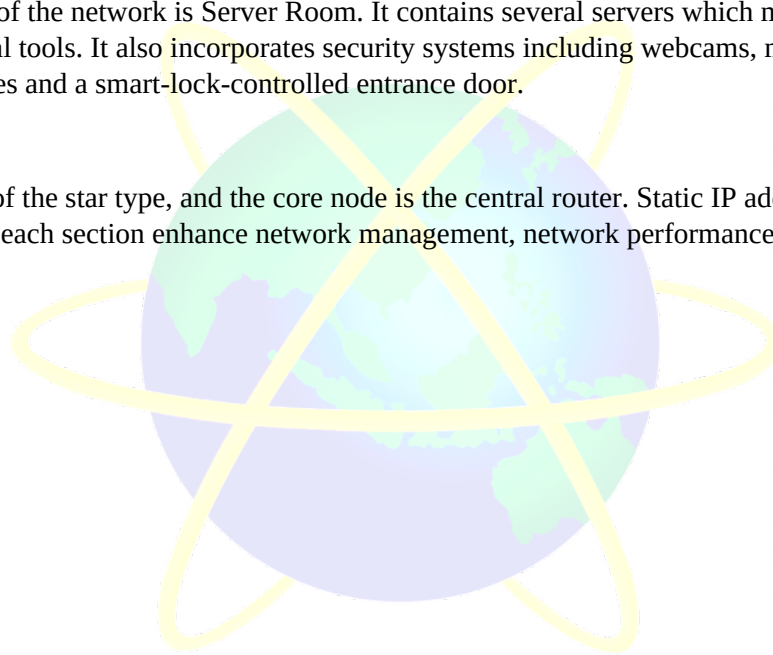
The network is well planned to facilitate centralized communications, secure segmentation and integration of smart devices. All the switches are linked to a central router which is responsible for routing between the subnets through static route, and each of the three zones are linked each to a separate switch.

The administrative center is referred to as the Staff Point. It consists of PCs and laptops to facilitate orders processing, personnel management and communication within the company. Other IoT activations in this area include available Bluetooth speakers and air conditioners.

The Guest Room is the front facing part. It has interactive gadgets like laptops to be used in customer services as well as an array of smart gadgets including fans, lights, humidifier, ACs among others to take care of customer comfort.

The technical core of the network is Server Room. It contains several servers which manage data, POS systems and internal tools. It also incorporates security systems including webcams, motion detectors, knee sensing devices and a smart-lock-controlled entrance door.

This is a topology of the star type, and the core node is the central router. Static IP addressing and separate subnets in each section enhance network management, network performance and security.



2.0 Student 4: KAMA L ASHRAF KAMAL

2.1 Topology Design (Kota Kinabalu)

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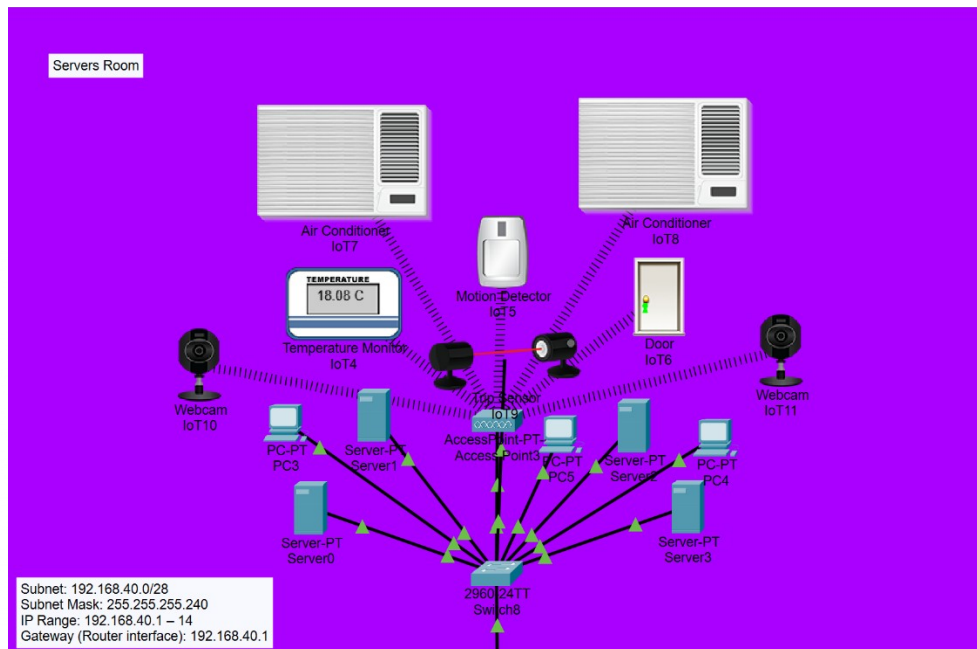


Figure 4.1.1 Kota Kinabalu

Subnet: 192.168.40.0/28

Subnet Mask: 255.255.255.240

IP Range: 192.168.40.1 – 14

Gateway (Router interface): 192.168.40.1

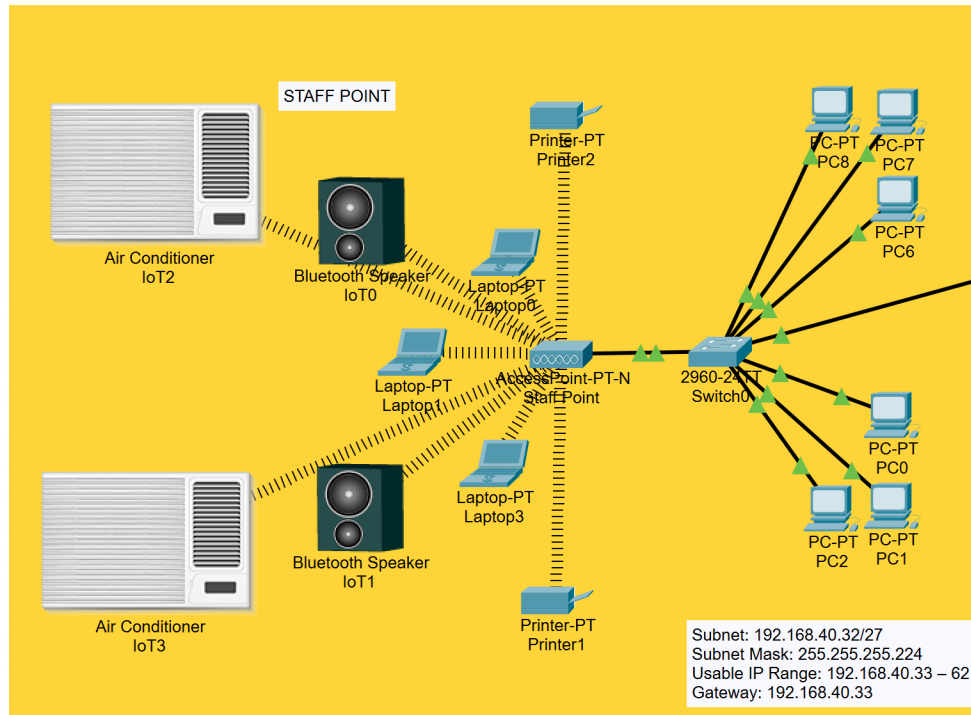


Figure 4.1.2 Kota Kinabalu

Subnet: 192.168.40.32/27

Subnet Mask: 255.255.255.224

Usable IP Range: 192.168.40.33 – 62

Gateway: 192.168.40.33

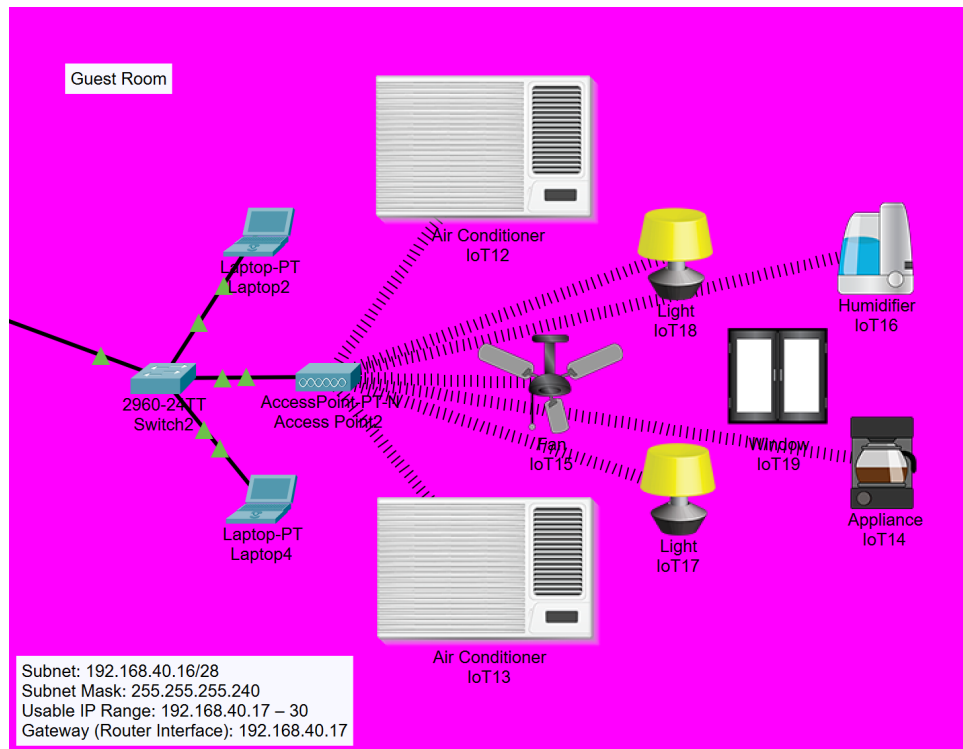


Figure 4.1.3 Kota Kinabalu

Subnet: 192.168.40.16/28

Subnet Mask: 255.255.255.240

Usable IP Range: 192.168.40.17 – 30

Gateway (Router Interface): 192.168.40.17

2.2 Topology Design – Introduction

The topology adopted in this network is that which is going to facilitate the activities of a company called SkyNetix. It has 4 branches in (Kuala Lumpur, Pulau Pinang, Johor Bahru, Kota Kinabalu) today we will focus on Kota Kinabalu that has three physical sections to it (Staff Point, Guest Room, Servers Room). The design is made to ensure:

- Central management by a central router
- Division of branches through switches
- Especially the utilization of end gadgets like PCs, laptops, servers, IoT devices (ACs and fans, cameras, etc.).

Its entire branches are hardwired and are maintained through the use of static routing where the performance is consistent and secure.

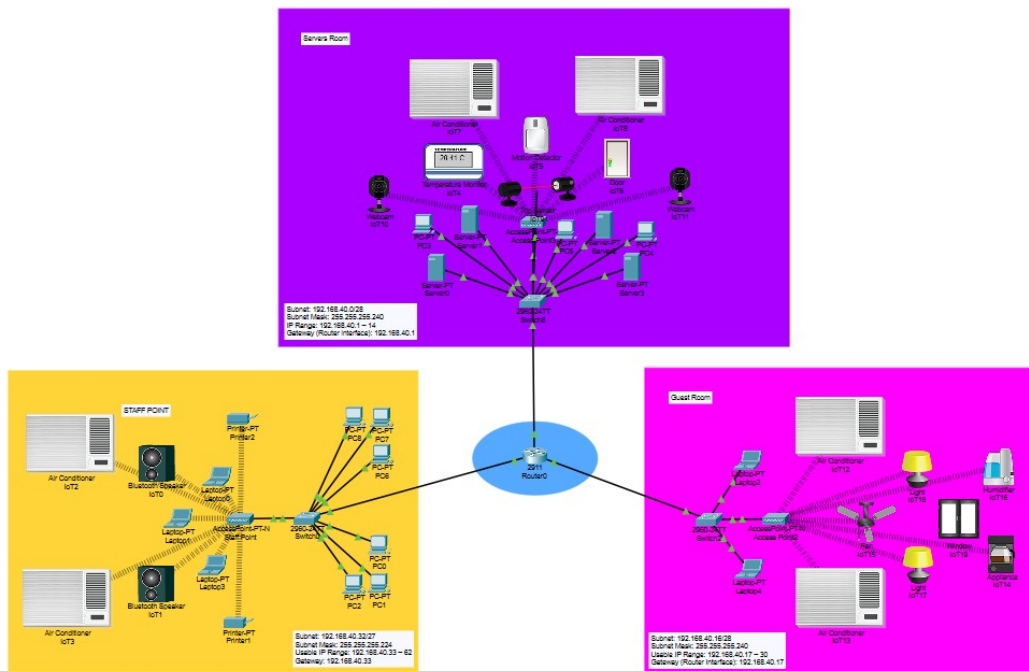


Figure 4.2.1 Kota Kinabalu

SkyNetix Kota Kinabalu branch is organized into three major areas, namely the Staff Point, the Guest Room, and the Server Room. Each zone is service to a certain range of business operational requirements and is said to be furnished with smart devices and computers to complement the working process, customer experience, and data management.

The administrative center of control is Staff Point. It has several PCs and laptops to help with the management of orders, staff and inventory and communication. There are also smart things, such as Bluetooth speakers and air conditioning, which are linked to enhancing the working environment.

The customer interface area is called The Guest Room. It has laptops to serve customers and smart devices like fans, lights, humidifiers and air conditioners to keep the guests in a modern and pleasant environment.

The main part of the network and security infrastructure is the Server Room. It has several servers on which it stores the data, point of sales devices, and internal communication systems. Moreover, the room has security-oriented equipment, such as webcams, motion sensors, trip sensors, and a smart access door, and laptops that could be used to track the activity in the room administratively or IT-wise.

Each of the zones connect to its switch and switches are connected to a central router. This configuration provides a clean topological boundary, good traffic management, scale and trouble shooting.



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2.3 Justification of Topology

Reason	Explanation
Centralized routing	All traffic flows through a central router, allowing easier monitoring.
Isolation between branches	Each branch uses its own subnet, improving security and reducing congestion.
Scalability	More branches can be added easily by connecting them to the central router.
Performance	Switch-based local networks offer high performance with minimal delay.
Cost-efficient hardware	Packet Tracer devices reflect typical real-world affordable networking gear.

STAR TOPOLOGY

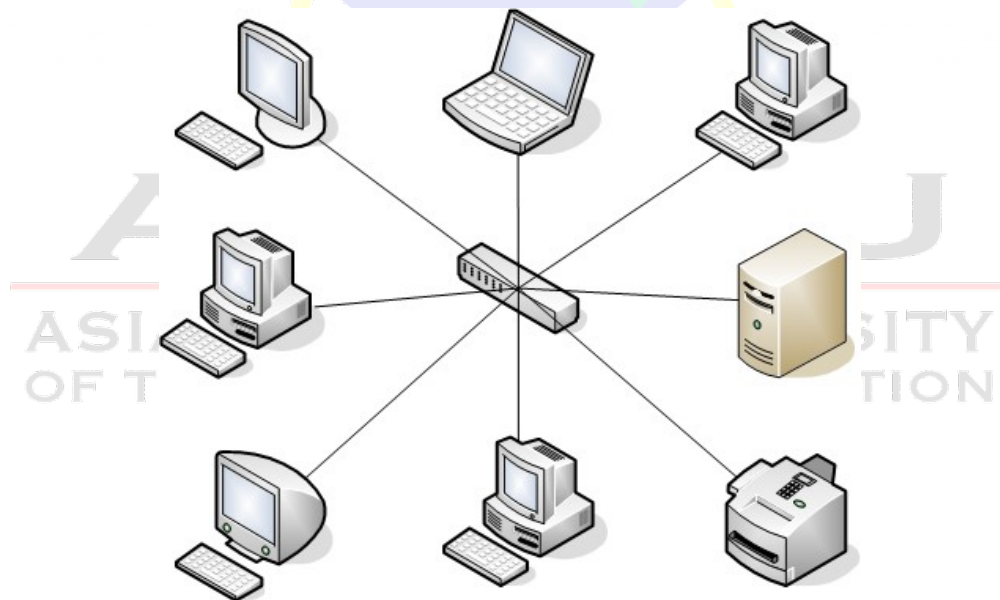
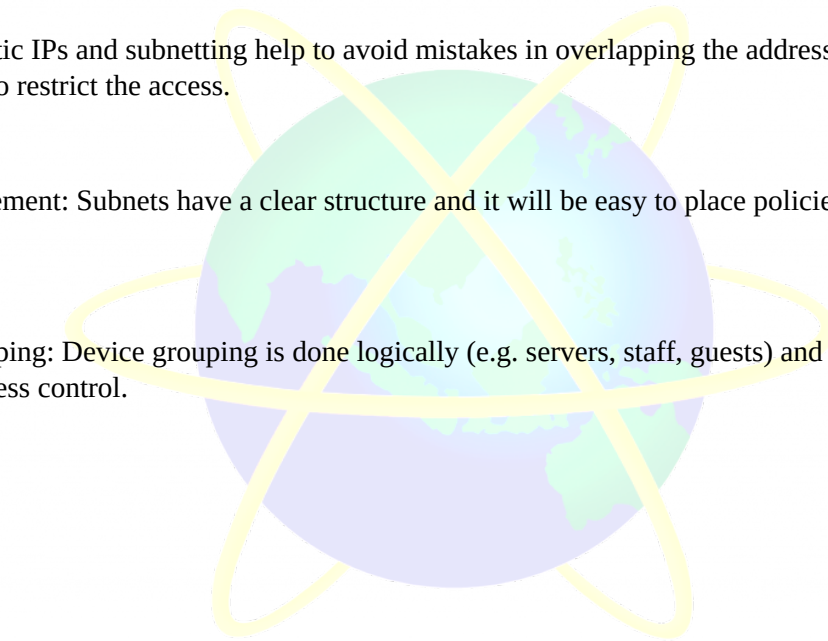


Figure 4.3.1 Kota kina

2.4 Advantages of the Topology Design

- **Simplicity of Troubleshooting:** The problems can be identified quickly with the help of centralized router and segmented networks.
- **Improved Bandwidth:** Bandwidth is utilized in a branch using the devices and switches thereby decreasing the load in the router.
- **Security:** Static IPs and subnetting help to avoid mistakes in overlapping the addresses and can be implemented to restrict the access.
- **Easy Management:** Subnets have a clear structure and it will be easy to place policies (such as ACLs) in future.
- **Device Grouping:** Device grouping is done logically (e.g. servers, staff, guests) and assists in offering role-based access control.



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2.5 Description

The full current configuration of every branch (3 Switches & router)

Router

```
interface GigabitEthernet0/0
  ip address 192.168.40.1 255.255.255.240
  duplex auto
  speed auto
!
interface GigabitEthernet0/1
  ip address 192.168.40.17 255.255.255.240
  duplex auto
  speed auto
!
interface GigabitEthernet0/2
  ip address 192.168.40.33 255.255.255.224
  duplex auto
  speed auto
!
interface Vlan1
  no ip address
  shutdown
```

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```
192.168.40.0/24 is variably subnetted, 6 subnets, 3 masks
C    192.168.40.0/28 is directly connected, GigabitEthernet0/0
L    192.168.40.1/32 is directly connected, GigabitEthernet0/0
C    192.168.40.16/28 is directly connected, GigabitEthernet0/1
L    192.168.40.17/32 is directly connected, GigabitEthernet0/1
C    192.168.40.32/27 is directly connected, GigabitEthernet0/2
L    192.168.40.33/32 is directly connected, GigabitEthernet0/2
```

Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	192.168.40.1	YES	manual	up
GigabitEthernet0/1	192.168.40.17	YES	manual	up
GigabitEthernet0/2	192.168.40.33	YES	manual	up
Vlan1	unassigned	YES	unset	administratively down

Switch 0

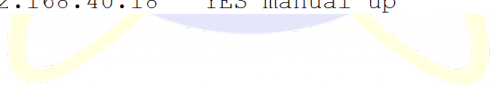
Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	up	up
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	up	up
FastEthernet0/11	unassigned	YES	manual	up	up
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	192.168.40.2	YES	manual	up	up

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2

Switch 2

Switch#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	192.168.40.18	YES	manual	up	up



VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2

Switch 8

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	up	up
FastEthernet0/7	unassigned	YES	manual	up	up
FastEthernet0/8	unassigned	YES	manual	up	up
FastEthernet0/9	unassigned	YES	manual	up	up
FastEthernet0/10	unassigned	YES	manual	up	up
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	192.168.40.33	YES	manual	up	up

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2

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2.6 IP ADDRESS SCHEME

Branch	Device	Interface	Network Address	IP Address	Usable Address Range	Broadcast Address	Default Gateway

Staff Point	PC 1	Gig0/0	192.168.40.0	192.168.40.2	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	PC 2	Gig0/0	192.168.40.0	192.168.40.3	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	PC 3	Gig0/0	192.168.40.0	192.168.40.4	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	PC 4	Gig0/0	192.168.40.0	192.168.40.5	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	PC 5	Gig0/0	192.168.40.0	192.168.40.6	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	PC 6	Gig0/0	192.168.40.0	192.168.40.7	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	Laptop 1	Gig0/0	192.168.40.0	192.168.40.8	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	Laptop 2	Gig0/0	192.168.40.0	192.168.40.9	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	Laptop 3	Gig0/0	192.168.40.0	192.168.40.10	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	Bluetooth Speaker 1	Gig0/0	192.168.40.0	192.168.40.11	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	Bluetooth Speaker 2	Gig0/0	192.168.40.0	192.168.40.12	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	AC 1	Gig0/0	192.168.40.0	192.168.40.13	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Staff Point	AC 2	Gig0/0	192.168.40.0	192.168.40.14	192.168.40.2 - .145	192.168.40.15	192.168.40.1
Guest Room	Laptop 1	Gig0/1	192.168.40.16	192.168.40.18	192.168.40.18 - .301	192.168.40.31	192.168.40.17
Guest	Laptop	Gig0/1	192.168.40.16	192.168.40.18	192.168.40.18 - .301	192.168.40.31	192.168.40.17

Room	2		.16	0.19	0.18 - .30	1	.17
Guest Room	Fan	Gig0/1	192.168.40 .16	192.168.4 0.20	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Guest Room	Light 1	Gig0/1	192.168.40 .16	192.168.4 0.21	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Guest Room	Light 2	Gig0/1	192.168.40 .16	192.168.4 0.22	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Guest Room	Humidifier	Gig0/1	192.168.40 .16	192.168.4 0.23	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Guest Room	Appliance	Gig0/1	192.168.40 .16	192.168.4 0.24	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Guest Room	AC 1	Gig0/1	192.168.40 .16	192.168.4 0.25	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Guest Room	AC 2	Gig0/1	192.168.40 .16	192.168.4 0.26	192.168.4 0.18 - .30	192.168.40.3 1	192.168.40 .17
Servers Room	Server 1	Gig0/2	192.168.40 .32	192.168.4 0.34	192.168.4 0.34 - .62	192.168.40.6 3	192.168.40 .33
Servers Room	Server 2	Gig0/2	192.168.40 .32	192.168.4 0.35	192.168.4 0.34 - .62	192.168.40.6 3	192.168.40 .33
Servers Room	Server 3	Gig0/2	192.168.40 .32	192.168.4 0.36	192.168.4 0.34 - .62	192.168.40.6 3	192.168.40 .33
Servers Room	Server 4	Gig0/2	192.168.40 .32	192.168.4 0.37	192.168.4 0.34 - .62	192.168.40.6 3	192.168.40 .33
Servers Room	Motion Detecto	Gig0/2	192.168.40 .32	192.168.4 0.38	192.168.4 0.34 - .62	192.168.40.6 3	192.168.40 .33

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Servers Room	Trip Sensor	Gig0/2	192.168.40.32	192.168.40.39	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	Webcam 1	Gig0/2	192.168.40.32	192.168.40.40	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	Webcam 2	Gig0/2	192.168.40.32	192.168.40.41	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	Laptop 1	Gig0/2	192.168.40.32	192.168.40.42	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	Laptop 2	Gig0/2	192.168.40.32	192.168.40.43	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	Laptop 3	Gig0/2	192.168.40.32	192.168.40.44	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	Door	Gig0/2	192.168.40.32	192.168.40.45	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	AC 1	Gig0/2	192.168.40.32	192.168.40.46	192.168.40.34 - .62	192.168.40.63	192.168.40.33
Servers Room	AC 2	Gig0/2	192.168.40.32	192.168.40.47	192.168.40.34 - .62	192.168.40.63	192.168.40.33

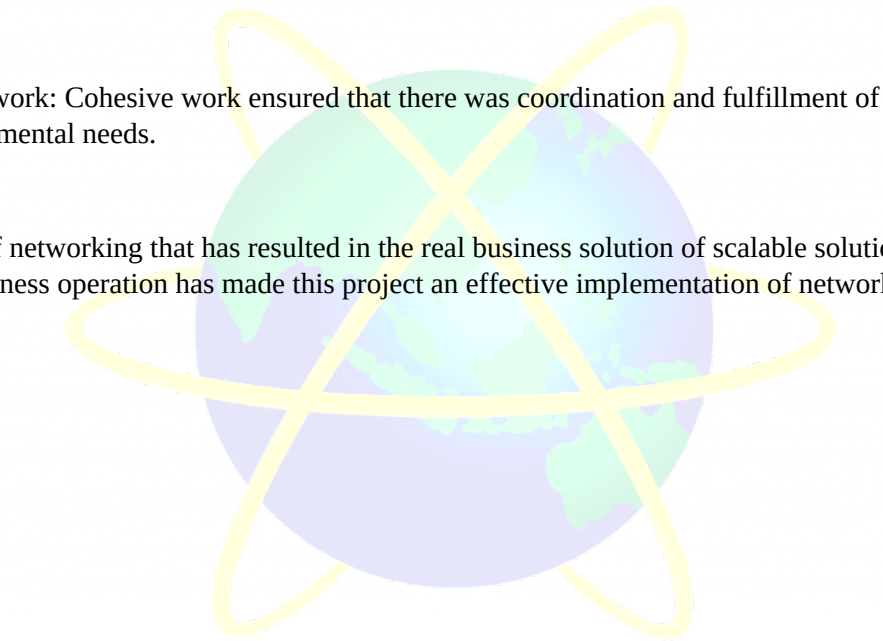
Conclusion

Subproject IT201 Introduction to Networking (CT042-3-1) has developed a highly organized network environment of CodeLink Solutions with its branches in Kuala Lumpur, Penang, Johor Bahru and Kota Kinabalu. The developments could be summarised as being:

- Topology: Star topology has been chosen because of reliability, expandability and easy to manage.

- Introduction IP Addressing involved the implementation of VLSM through effective subnetting so as to utilize IP address per department.
- IoT Implementation: More automation and security levels were enhanced due to the presence of intelligent equipment (RFID, sensors, security cameras).
- Testing: Functional testing and connectivity of all the network segments was done.
- Teamwork: Cohesive work ensured that there was coordination and fulfillment of the departmental needs.

The concept of networking that has resulted in the real business solution of scalable solutions and security to the real business operation has made this project an effective implementation of networking concepts.



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References :

- Roman, R., Zhou, J., & Lopez, J. (2021). *On the features and challenges of security and privacy in distributed internet of things*. Computer Networks, 192, 108075. <https://doi.org/10.1016/j.comnet.2021.108075>
- NIST. (2022). *Security considerations for IoT device manufacturing (NISTIR 8259B)*. National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.IR.8259B>

Norman, T. L. (2021). *Electronic Access Control* (3rd ed.). Butterworth-Heinemann.

Garcia, M. L. (2020). *The design and evaluation of physical protection systems* (3rd ed.). Butterworth-Heinemann.

Cisco. (2023). *Catalyst switch configuration guides*. <https://www.cisco.com/c/en/us/support/docs/switches/catalyst-9000-series-switches>

Cisco IoT. (2023). *Cisco IoT security best practices*. <https://www.cisco.com/c/en/us/solutions/internet-of-things/iot-security.html>

ASHRAE. (2021). *Thermal guidelines for data processing environments* (5th ed.). American Society of Heating, Refrigerating and Air-Conditioning Engineers.

Uptime Institute. (2022). *Data center site infrastructure tier standard: Topology*. <https://uptimeinstitute.com>

Al-Fuqaha, A., et al. (2021). *Internet of Things: A survey on enabling technologies, protocols, and applications*. IEEE Communications Surveys & Tutorials, 23(3), 2049-2107.

RFC 1918. (1996). *Address allocation for private internets*. IETF. <https://tools.ietf.org/html/rfc1918>

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